A UNIFIED SMART PLATFORM FOR ROADSIDE ASSISTANCE AND AUTOMOTIVE SERVICES IN SRI LANKA

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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Abstract

Kaduna.lk is a roadside-assistance platform for Sri Lanka. This thesis presents its Geo-Intelligence component: an interpretable five-factor score (1–10) of a single incident's traffic impact, a congestion estimator and a hotspot analysis, deployed as a live, tested service that the dispatch component calls. A senior Department of Motor Traffic (DMT) official stated that the DMT keeps no timed, location-coded incident records. As none could be obtained, evaluation used simulation and expert judgement, with 500 synthetic incidents on the real Colombo road network. Against a 120-scenario SUMO (Simulation of Urban MObility) grid, the score ranked simulated incidents moderately (Spearman $\rho = 0.67$; Pearson $r = 0.60$, missing the $r \geq 0.75$ target); the grid could identify only lanes blocked, which drove the simulated slowdown. In an internal-consistency check with demand generated from the model's own profile, the score ranked total delay closely ($\rho = 0.91$). Density-based clustering found 25 hotspots: high-impact incidents fell inside them more often than other incidents (odds ratio 2.3), and ten hotspot centres lay within 1.5 km of named blackspots, against about five expected by chance ($p \approx 0.02$). Hypothesis H2, that impact-first dispatch reduces delay, was rejected: it was worse than nearest-first at realistic fleet load (19.1% at 0.88 utilisation) and under overload (23.7–64.7%). On a real Colombo sub-network, diverting all traffic around a partially blocked arterial increased network total travel time by 18.8 vehicle-hours per scenario in the 40-minute window. The DMT official, the single structured respondent, rated usability at 80.0 on the System Usability Scale (indicative only); the scenario-ranking comparison was inconclusive ($\rho = -0.05$). The score is recommended for triage and exploratory unit placement, not for queue ordering, and the platform could become a source of the missing incident records.

Keywords: traffic incident impact, microscopic traffic simulation, spatial hotspot analysis, roadside assistance, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
ATMS	Advanced traffic management system
B2B	Business-to-business
B2G	Business-to-government
C4	Context, Containers, Components and Code (architecture model)
CLF	Capacity loss factor
CSV	Comma-separated values
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DIG	Deputy Inspector General (of Police)
DMT	Department of Motor Traffic
FHWA	Federal Highway Administration (United States)
FR	Functional requirement
Gi*	Getis-Ord local concentration statistic
GPS	Global Positioning System
H1, H2	Hypotheses 1 and 2
HCM	Highway Capacity Manual
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IS	Impact score
ISF	Incident severity factor
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
JWT	JSON Web Token
KDE	Kernel density estimation
LF	Location factor
LWR	Lighthill–Whitham–Richards (kinematic wave model)
MAE	Mean absolute error
NFR	Non-functional requirement
NTC	National Transport Commission
OSM	OpenStreetMap

Abbreviation	Meaning
OWASP	Open Worldwide Application Security Project
p95	95th percentile
PEP	Python Enhancement Proposal
PP1	First progress presentation
REST	Representational State Transfer
RQ	Research question
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SLSQP	Sequential Least Squares Programming
SO	Specific objective
SUMO	Simulation of Urban MObility
SUS	System Usability Scale
TF	Temporal factor
TIM	Traffic incident management
TVF	Traffic volume factor
VHL	Vehicle-hours lost
XML	Extensible Markup Language
YAML	YAML Ain't Markup Language

1 INTRODUCTION

1.1 Overall Project Context

Road traffic in Sri Lanka is concentrated in and around Colombo. In the most recent open district-level police tables (2012) [1], the Colombo District accounted for 16,549 of 61,587 recorded road accidents (26.87%). Road traffic injury remains a leading cause of death worldwide, with most of the burden in low- and middle-income countries [2]. Every breakdown or collision on an urban road also removes capacity while the vehicle remains, delaying many uninvolved road users.

Breakdown help in Sri Lanka is still arranged mainly by telephone, through garages, towing operators, insurers and motoring associations. The Police emergency number 119 and the 1990 Suwa Seriya ambulance service [3] deal with crime and medical care, not with clearing disabled vehicles. A retired Deputy Inspector General (DIG) of Police (the ex-DIG), interviewed for this study, described a formal response pathway (hotline, police attendance, private towing) that operates on the expressways only. For ordinary roads, the ex-DIG stated, "there is no similar system to help locals" (personal communication, 27 August 2026).

Kaduna.lk (group R26-SE-026) addresses this fragmentation with one platform built from four research components: triage and dispatch (Janukshan, IT22635266); geo-intelligence and traffic impact analysis (M. M. Asath, IT22633422), the subject of this thesis; predictive maintenance (Herath D M S T, IT22639776); and insurance claims and privacy (Dilnuk De Silva, IT22001252). The platform comprises an Expo mobile application, a Next.js web application, Supabase authentication with JSON Web Tokens (JWT), PostgreSQL and one back-end service per component [4]. In its working path, the mobile application submits a request to dispatch, dispatch requests an impact score from the geo-intelligence service, the score informs triage, and a provider is chosen.

1.2 Individual Component Context

The geo-intelligence component answers one question for the platform: "what is this incident doing to the road?" It does not decide who responds. Geo-intelligence acts

like a traffic map, dispatch like a ride-matching service. Dispatch consumes the score, never the reverse.

The component delivers four things:

1. An **impact score** from 1 to 10 for a single incident, built from five factors: capacity loss (CLF, from lanes blocked), traffic volume (TVF), a temporal factor (TF, time of day and day of week), a location factor (LF, road class) and an incident severity factor (ISF, incident type). The score maps to four priority bands: LOW, MEDIUM, HIGH and CRITICAL.
2. **Congestion estimates**: queue length, vehicle-hours lost (VHL) and recovery time.
3. **Hotspot analysis**: spatial clusters of incidents in Colombo, found with kernel density estimation (KDE) and Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and ranked by the impact of the incidents they contain.
4. **A rerouting analysis**: an analytical and a microsimulation evaluation of the effect of diverting traffic around a blocked road segment.

The score, congestion estimates and hotspots are served by the production FastAPI service and shown on the operations dashboard; simulation and validation run in a separate research pipeline (Appendix C). The service supports operator triage, a bounded term in dispatch's provider cost and exploratory unit-placement rankings. It does not reorder the dispatch queue or offer detour advice. The primary users are operations-desk staff and dispatch as a machine consumer; drivers benefit indirectly, while the Police and Department of Motor Traffic (DMT) are prospective users of the impact and hotspot views.

1.3 Problem Statement

When several requests are open at once, the platform must decide which incident is most urgent, yet without traffic information every request looks the same. A breakdown blocking one lane of Galle Road at 18:00 on a weekday and a flat tyre on an empty residential lane at 03:00 arrive as identical records, although the first may delay thousands of vehicles and the second almost nobody. Authorities face a similar

blindness at network level: they know where accidents happen, but have no repeatable view of where incident impact concentrates.

The data normally used to build and test an impact model could not be obtained. A senior DMT official stated that the DMT keeps no timed, location-coded incident records, that it receives only major or high-priority incidents, and that all others are handled by the Police (personal communication, 5 October 2026). Located accident data exist only as aggregates, such as the National Transport Commission (NTC) list of accident-prone locations. An impact model must therefore be built from traffic-flow theory and published parameters, and evaluated by simulation and expert judgement.

The problem addressed by this thesis is therefore: **there is no interpretable, quick-to-compute measure of how much a single reported incident will disrupt traffic on Sri Lankan urban roads, and no recomputable view of where high-impact incidents cluster, that a roadside-assistance platform and its operations staff can act on.**

1.4 Research Gap

Four gaps, developed in Chapter 2, motivate the study.

Gap 1: no incident impact score that can be built and graded without incident logs. Incident impact has been studied with microscopic simulation on United Kingdom motorways [5], with machine-learning severity classification [6] and with clearance-time prediction [7]. These approaches either produce no reusable per-incident score or require observed disruption data, which could not be obtained in Sri Lanka.

Gap 2: hotspot analysis in Sri Lanka is static and frequency-based. The ex-DIG described a police study that identified accident-prone locations, but as a one-off manual exercise that has not been recomputed (personal communication, 27 August 2026). The gap is its staleness and its focus on counts rather than traffic impact; the methods themselves, such as density-based clustering [8], KDE [9] and Getis-Ord statistics [10], are well established.

Gap 3: impact estimation and incident response are studied separately. Traffic engineering studies how incidents create congestion [11], [12]; operations research studies responder allocation [13], [14]. Whether an impact signal improves response, by reordering dispatch or by diverting traffic, was not found to have been tested for roadside assistance in the sources reviewed.

Gap 4: validation where incident records cannot be obtained. Published impact models are trained or calibrated on detector data or incident logs. Where these cannot be obtained, as in this study, validation has to combine other independent sources of evidence.

1.5 Aim and Objectives

Aim. To design, build and evaluate a geo-intelligence service that estimates the traffic impact of individual roadside incidents in Colombo, identifies where high-impact incidents concentrate, and determines how this intelligence can best support triage, rerouting and dispatch.

The proposal expected priority-based dispatch to reduce congestion; this thesis tests that outcome as hypothesis H2. The seven specific objectives (SO) are:

1. **SO1:** develop and deploy an interpretable five-factor score with a 500 ms response target.
2. **SO2:** evaluate the score against SUMO microsimulation, including factor, robustness and transfer analyses.
3. **SO3:** identify impact-ranked hotspots and compare them with spatial statistics and known blackspots.
4. **SO4:** estimate queue length, vehicle-hours lost (VHL) and recovery time, and evaluate ranking and magnitude.
5. **SO5:** build the operations views and integrate the score into dispatch.
6. **SO6:** test impact-ordered dispatch against nearest-first and evaluate rerouting.
7. **SO7:** obtain structured expert feedback on relevance, usability, reliability and scenario rankings.

1.6 Research Questions and Hypotheses

Four research questions (RQ) guide the study:

- **RQ1.** How closely does an interpretable weighted impact score agree with incident congestion produced by microscopic traffic simulation, and which factors carry that agreement?
- **RQ2.** Do incidents in Colombo form spatially coherent clusters, and how do these clusters relate to known accident blackspots?
- **RQ3.** Does ordering dispatch by impact reduce total delay, and if not, what is a more effective use of impact information?
- **RQ4.** Do domain experts regard the factors, outputs and rankings of the model as relevant, usable and credible?

RQ1 and RQ3 are tested through two hypotheses (H), fixed before the experiments were run:

- **H1 (score validity).** The deployed score correlates with SUMO-simulated speed reduction with Pearson $r \geq 0.75$ across the scenario grid.
- **H2 (impact-ordered dispatch).** Ordering dispatch by impact reduces total network delay compared with nearest-first dispatch, measured with a cost that is not derived from the score. The proposal set a target reduction of at least 20%.

A cost computed from the score would reward the score by construction. In the experiment, total delay is the incident-edge VHL predicted by a cost model fitted to simulation (Section 3.3). RQ2 and RQ4 are answered with target-based metrics and descriptive evidence.

1.7 Scope, Limitations and Assumptions

The study covers 500 synthetic incidents on the real Colombo OpenStreetMap (OSM) network, a 120-scenario SUMO corridor grid, a real Colombo sub-network and two stakeholder sessions. The deployed weights are theory- and literature-based priors; simulation-fitted weights are sensitivity-only, and the sensitive-location overlay sits outside the evaluated five-factor score. Road attributes come from OSM, while capacities and clearance times are literature-based defaults. Provider

assignment, teammates' components and integration with government systems are outside scope. Section 7.5 consolidates the threats to validity.

1.8 Individual Contribution

The author designed the model and incident generator; conducted the simulation, hotspot, rerouting, dispatch and stakeholder studies; and built, tested and deployed the geo-intelligence service and operations-dashboard views. The author also defined and implemented dispatch's score client and Colombo-time handling. Janukshan built the dispatch service and optimiser; authentication, the mobile application, web shell, live incident feed and hosting were team work. The research contributions—an interpretable triage score under data scarcity, a characterised negative dispatch result and a multi-source validation protocol—are assessed in Section 7.3.

1.9 Thesis Structure

Chapter 2 reviews related work; Chapters 3–5 describe methodology, design and implementation; Chapters 6–7 present and interpret the evidence; and Chapters 8–9 address commercialisation and conclusions. This nine-chapter structure follows the official SLIIT individual thesis template; relative to the general dissertation outline in the SLIIT documentation guidelines, the background literature and research gap appear in Chapters 1–2, methodology, testing and implementation in Chapters 3–5, results, findings and discussion in Chapters 6–7, commercialisation aspects in Chapter 8 and the individual contribution in Section 1.8. The appendices contain the instruments, artificial-intelligence disclosure, artefact inventory and traceability matrix (Table D.1).

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This review is confined to the component's own domain: traffic flow, incident impact and duration, spatial hotspot analysis, multi-criteria scoring and the use of traffic information in incident response. Dispatch algorithms, fault prediction, insurance and data-protection law belong to teammates' theses.

Sources were found in IEEE Xplore, SpringerLink, ScienceDirect, MDPI and Google Scholar, combining "traffic incident" or "vehicle breakdown" with terms such as "impact", "capacity reduction", "clearance time", "SUMO", "rerouting", "hotspot", "DBSCAN" and "incident response". Recent work (2018–2025) was preferred for applied methods, and foundational works were kept where they define a method still in use. The Highway Capacity Manual (HCM) [15] and the Federal Highway Administration (FHWA) Traffic Incident Management (TIM) Handbook [16] were treated as authoritative for engineering practice.

2.2 Domain and Theoretical Foundations

2.2.1 Capacity, demand and queueing at incidents

The capacity of a road is the maximum sustained flow it can carry. The HCM gives standard capacity values per lane by facility type and treats incidents as events that temporarily remove part of that capacity [15]. A simple view gives each lane an equal share, so blocking one lane of two would remove half. The HCM does not take this view: it tabulates the proportion of capacity that remains under incident conditions by lanes blocked and total lanes. With one lane blocked, 0.35, 0.49 and 0.58 of capacity remain on two-, three- and four-lane freeway segments, well below the lane shares of 0.50, 0.67 and 0.75 [15]. Empirical work agrees: Knoop, Hoogendoorn and van Zuylen observed two motorway accidents from a helicopter and found the outflow of the remaining lanes about 50% below their free-flow capacity, with capacity in the unblocked opposite direction also reduced by rubbernecking (drivers slowing to look at the incident) [17]. A lane-share measure therefore understates capacity loss and so underestimates impact; it is a lower bound on the factor, not a conservative choice for triage.

Congestion forms when demand q exceeds the capacity left by the incident. The deterministic input-output (cumulative-count) queue describes the result, and Newell showed that cumulative counts give an exact way to analyse such queues [18]. For an incident of duration T that reduces capacity from c to μ, after which full capacity is restored, the total delay is the area between the arrival and departure curves:

$$D = (T^2 (q - \mu)(c - \mu))/(2 (c - q)) \quad (2.1)$$

Two properties matter for the component. Delay is zero while demand q stays below the remaining capacity μ, so capacity loss and demand are jointly necessary. And delay grows with the square of duration, so shortening an incident has a more than proportional effect. These properties are the basis of the capacity loss (CLF), traffic volume (TVF) and incident severity (ISF) factors.

2.2.2 Kinematic waves and shockwaves

The input-output model treats the queue as a point. The Lighthill–Whitham–Richards (LWR) model instead represents traffic through conservation of vehicles [19], [20]:

$$(\partial k)/(\partial t) + (\partial q)/(\partial x) = 0 \quad (2.2)$$

where k is density and q is flow. Discontinuities between traffic states propagate as shockwaves, determining how queues spread upstream; Daganzo's cell transmission model made this theory computable on networks [21]. The component uses the simpler cumulative-count model and converts queued vehicles to distance through jam density. Because it has no calibrated fundamental diagram, its outputs are tested against simulation rather than presented as an LWR implementation.

2.2.3 Incident duration

Because delay rises with the square of duration, expected duration is central to impact. The Traffic Incident Management Handbook divides the timeline into detection, verification, response, clearance and recovery [16]. Duration models depend on incident and roadway characteristics but require timed logs [22], [23]. In their absence, this study uses literature-based defaults by incident type and compares them with expert judgement.

2.2.4 Spatial hotspot methods

Road-safety agencies identify hazardous locations ("blackspots") to focus treatment, and Elvik found that European definitions differ in, among other things, whether severity is taken into account [24]. Three families of spatial methods are relevant.

Kernel density estimation (KDE) sums a smooth kernel over each event to give a continuous density surface, and the bandwidth controls the result more than the kernel [25]. Zheng et al. showed that bandwidth selection strongly affects which hotspots are found [9], and Okabe, Satoh and Sugihara constrained the kernel to the road network [26].

Density-based clustering. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) groups points that have at least a minimum number of neighbours within a radius ε and labels the rest as noise [8]. It needs no preset number of clusters and leaves isolated events unclustered, which suits incident data. Cluster quality can be measured with the silhouette coefficient [27].

Spatial autocorrelation statistics test whether a pattern could have arisen by chance. Moran's I measures global autocorrelation [28], and the Getis-Ord statistics measure local concentration [29]; the Gi* variant identifies locations whose neighbourhoods are significantly high (local "hot spots", distinct from the clusters called hotspots in this study) [30]. Because Gi* is computed at every location, its significance needs correction for multiple testing.

2.2.5 Multi-criteria weighted scoring

The impact score (IS) combines several factors into one number. Multi-attribute value theory shows that, under preferential independence, preferences over several attributes can be represented by an additive value function [31]:

$$IS = \Sigma_i \, w_i \, F_i, \quad \Sigma_i \, w_i = 1, \quad 0 \leq F_i \leq 1 \quad (2.3)$$

Weights can be set directly as expert or theory-based priors, derived from pairwise comparisons in the Analytic Hierarchy Process [32], or fitted where outcome data exist. In this study they were set as priors (Section 4.4.1). An additive score suits operations because each factor's contribution can be shown, it needs no training data, and it can be checked factor by factor. Rudin argues that interpretable models of this

kind should be preferred to black boxes for high-stakes decisions [33]. Its weaknesses are that additivity ignores interactions (capacity loss matters only when demand is high), that factors sharing inputs are not independent attributes, that prior weights may not match each factor's influence, and that the scale is ordinal until calibrated. The alternative is to score a monotone transform of the delay of Equation 2.1 directly; that is dimensionally meaningful but inherits every error in the capacity and demand inputs, whereas an additive score shows each factor's contribution and degrades more gracefully when one input is wrong. The validation must therefore test whether the score ranks incidents as an independent congestion measure does, and which factors carry that agreement.

2.3 Related Research

2.3.1 Incident impact estimation

Koliou, Quddus and Michalaki used VISSIM microsimulation with real incident data from a United Kingdom motorway to estimate the effect of breakdowns under different connected-vehicle penetration rates, and found that earlier detection and warning reduced both delay and conflicts [5]. It is the closest in purpose to this study but does not produce a reusable per-incident score. Ara and Hashemi classified the severity of accident impact on traffic flow with an ensemble of machine-learning models [6]; their approach learns from observed flow disruption and so needs such observations. Kwon, Mauch and Varaiya showed that incidents form a distinct, measurable share of freeway delay [34], which supports estimating incident delay separately from recurrent congestion.

2.3.2 Clearance-time prediction

Tang et al. predicted incident clearance time with Extreme Gradient Boosting and found non-linear effects of incident, roadway and environmental factors [7]. With the hazard-based models of Nam and Mannering [22] and the review by Li et al. [23], this work establishes incident type and lanes affected as informative about duration, which supports the use of incident type in the severity factor.

2.3.3 Microscopic simulation of incidents

Simulation of Urban MObility (SUMO) is an open-source microscopic traffic simulator with network import, demand generation, runtime control through its Traffic Control Interface and per-edge measurements [35]. Smith, Djahel and Murphy used SUMO to evaluate the effect of road incidents on congestion and added runtime rerouting that lets vehicles bypass blocked roads, which reduced travel times [11]. T. Li et al. built a SUMO extension that generates incidents and dispatches emergency vehicles on a synthetic grid network [12].

These studies establish SUMO as a credible instrument for incident experiments. None uses simulation to grade a separate closed-form impact score, which is its role in this study. They also leave the network side of diversion open. Applying the queueing argument of Section 2.2.1 to the detour shows why: a diverted vehicle avoids the queue but pays the extra detour distance, and the detour has finite capacity, so diverted flow can congest it in turn. The net network benefit of rerouting therefore depends on detour length, spare detour capacity and the share of drivers who divert, which is the question examined in Chapter 6.

2.3.4 Hotspot detection for road incidents

Applied hotspot studies mostly use crash records. Anderson combined KDE with K-means profiling for London [36], and Zhang et al. combined Getis-Ord Gi* with DBSCAN and argued that the two together are more robust than either alone [10]. Boeing's OSMnx library made it routine to relate incident locations to road class, lane count and network structure from OpenStreetMap (OSM) [37], [38]. In this body of work hotspots are defined by accident frequency, sometimes weighted by injury severity, rather than by traffic disruption, and the analysis is usually a one-off study of a historical dataset.

2.3.5 Incident response logistics

Zografos, Androutsopoulos and Vasilakis developed a real-time decision support system for incident response logistics [13]. Buil, de Armas, Riera and Orozco allocated roadside-assistance resources in real time for a roadside-assistance company, minimising operational cost and arrival time with heuristics and linear programming [14]; priority there comes from service-level agreements, not from

traffic impact. The ambulance-dispatch literature adds a caution. Jagtenberg, Bhulai and van der Mei showed that sending the closest idle unit is not always optimal, but that the gains from deviating depend on the objective and on travel times and future coverage [39]. Closest-first is therefore a strong baseline, and any priority rule must be tested against it. Hypothesis H2 is designed as that test.

2.4 Existing Systems, Products and Techniques

Navigation and traffic platforms. Google Maps shows live speeds and reported incidents and routes around congestion; Waze relies on drivers to report crashes and stopped vehicles. Amin-Naseri et al. found that Waze reports covered 43.2% of the records of an advanced traffic management system (ATMS) and arrived 9.8 minutes earlier on average than a probe-based alternative [40]. TomTom's flow service returned no data for Colombo in this study, whereas Google's Routes service covers the city. These platforms report congestion once it appears in speed data; they do not explain why one incident is more disruptive than another or offer a per-incident score to a dispatcher. They complement the component and could supply live speeds.

Traffic incident management programmes. In countries with mature TIM programmes, traffic management centres combine detectors, cameras, service patrols and multi-agency procedures, guided by documents such as the TIM Handbook [16]. They depend on detector networks and incident logs, which Sri Lankan urban roads lack, so the component works from road attributes and incident descriptions alone.

The Sri Lankan context. No published system that estimates the traffic impact of individual incidents in Sri Lanka was found. The retired Deputy Inspector General (DIG) of Police interviewed for this study (the ex-DIG) described a formal response pathway on the expressways only and the one-off blackspot study noted in Section 1.4. Open accident data are aggregated by police division and year, with no per-incident coordinates or times [1], [41], and the National Transport Commission (NTC) publishes accident-prone locations as a list. A senior official of the Department of Motor Traffic (DMT) stated that the department keeps no timed, location-coded incident records (personal communication, 5 October 2026). Per-

incident records could therefore not be obtained, which shapes the validation design of this study.

2.5 Gap Synthesis and Conceptual Framework

2.5.1 Comparison with related work

Table 2.1 compares representative related work with this study. "Partial" denotes an indirect or limited capability.

Table 2.1: Comparison of related work and existing systems with this study

Work or system	Incident impact	Simulation	Hotspots	Response/rerouting	Sri Lankan service
Koliou et al. [5]	Partial	Yes	No	Warning only	No
Ara and Hashemi [6]	Learned severity	No	No	No	No
Smith et al. [11]	Partial	Yes	No	Rerouting	No
Zheng and Zhang et al. [9], [10]	No	No	Count-based	No	No
Zografos and Buil et al. [13], [14]	No	No	No	Resource allocation	No
Google Maps and Waze	Live delay	No	No	Driver rerouting	Partial
This study	Interpretable score	Yes	Impact-ranked	Triage; rerouting experiment	Yes

The comparison supports the four gaps of Section 1.4: no per-incident impact score that can be built and graded without incident logs; count-based, static hotspot analysis; separate literatures on impact and response; and impact models built only where incident logs exist.

2.5.2 Conceptual framework

The framework follows one causal chain: an incident removes capacity; delay appears when demand exceeds what remains; and delay grows with duration. Capacity loss (CLF) represents lanes removed; traffic volume (TVF) and time (TF) represent demand; location (LF) represents road importance; and incident severity (ISF) represents expected duration. These factors form the 1–10 triage score, while hotspots, dispatch ordering and rerouting test distinct operational uses.

With no recorded ground truth, evaluation combines SUMO, transfer to a real road topology and structured expert feedback. These sources test different parts of the framework rather than forming a single claim of external validation.

3 RESEARCH METHODOLOGY

3.1 Research Design

This study follows design-science research [42], [43]: it builds an artefact, a service that scores the traffic impact of a roadside incident, and evaluates it against criteria fixed in advance.

The evaluation has two strands. The first compares the impact score with congestion in the Simulation of Urban MObility (SUMO) microsimulator [35] and runs rerouting and dispatch experiments on the real Colombo OpenStreetMap (OSM) network. The second is stakeholder validation with a retired Deputy Inspector General of Police (ex-DIG) and a senior official of the Department of Motor Traffic (DMT). Per-incident, timed, geocoded records were not obtainable for this study: the DMT official stated that the department keeps none, and the located accident data that exist, such as the National Transport Commission (NTC) accident-prone list, are aggregates (Sections 1.3 and 6.3.3). Simulation therefore supplies a controlled ground truth, and expert judgement checks the factors against local practice.

H1 expects the deployed score to correlate with SUMO congestion at Pearson $r \geq 0.75$ (research question RQ1). H2 expects impact-ordered dispatch, compared with nearest-first dispatch under a cost not derived from the score, to reduce vehicle-hours lost (VHL) by at least 20% (RQ3); in the experiment, VHL is the delay on each incident's edge as predicted by a cost model trained on SUMO (Section 3.3). RQ2 (hotspots) and RQ4 (expert view) use target-based metrics. Figure 3.1 shows the phases.

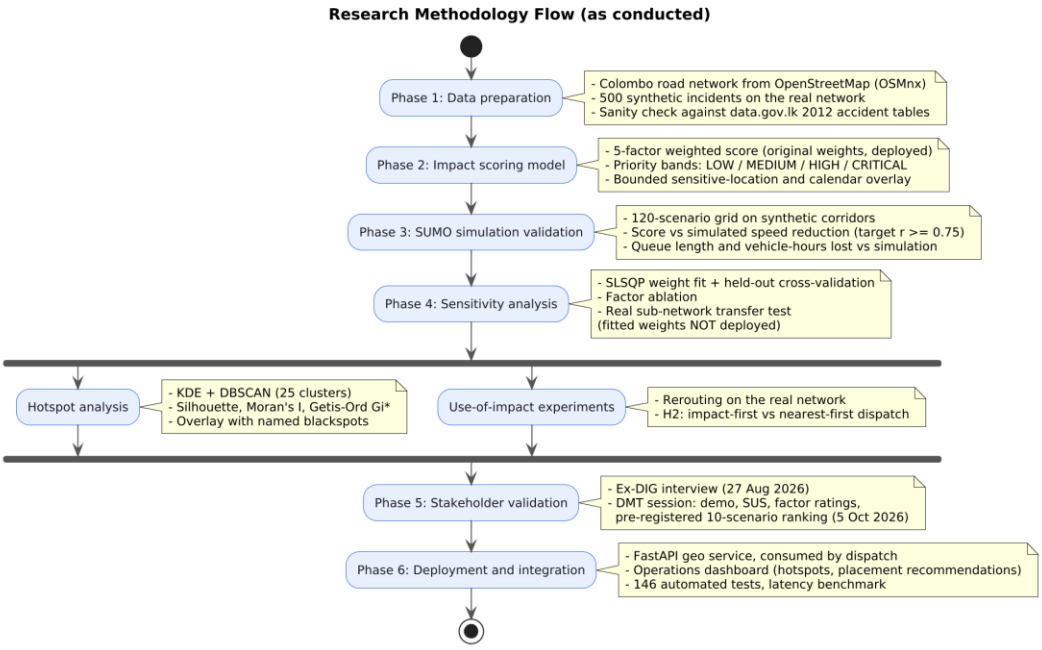


Figure 3.1: Research methodology flow, as conducted (original in colour)

3.2 Data, Participants and Study Materials

3.2.1 Road network and incidents

The study area is Colombo District. Its drivable network was downloaded from OSM [44] with OSMnx [37], [38], giving a directed graph of 42,989 nodes and 96,881 edges with length, road class and, where tagged, lane count and speed limit.

The model was developed on 500 synthetic incidents placed on this real geometry with a fixed seed. Each incident receives a road class from a prior that favours arterials (primary 0.25, trunk 0.20, secondary 0.20, tertiary 0.12, residential 0.10, motorway 0.08, others 0.05), then an edge in proportion to length, and one of seven incident types in the proportions of Table 4.2. The type sets the lanes blocked and a mean clearance time; the duration is drawn from a normal distribution about that mean with a 30% standard deviation (at least 10 minutes). The hour follows a curve with morning and evening peaks, and weekends are down-weighted.

3.2.2 Official statistics

The data.gov.lk accident table for 2012 [1] records 26.87% of national accidents in Colombo. Aggregated by district, it supports the choice of study area but cannot place incidents.

3.2.3 Simulation scenarios

The SUMO ground truth is a grid of 120 synthetic corridors of three 1 km edges, with the incident on the middle edge. Road class ranges from trunk to residential, lane count from two to four, all seven incident types appear, and demand is 0.5, 0.8, 1.1 or 1.4 times the model's road-class capacity. Each 75-minute run uses SUMO's default Krauss driver model, one fixed seed and an incident from minute 10; blocked lanes are slowed to 0.1 m/s and edge measures are collected every minute. The capacity convention keeps demand below the physical capacity of the simulated multi-lane corridor, so queues form mainly where blockage removes most lanes.

Two features limit what the grid can identify. Every major accident blocks two lanes and no other type does, so the grid cannot separate incident type from lanes blocked; among the 95 single-lane scenarios, incident type correlates with speed reduction at $r = -0.07$. Demand is also sampled independently of hour and road class, from which the traffic volume (TVF), temporal (TF) and location (LF) factors are computed, so these have no pathway to the outcome. The grid therefore identifies lanes blocked only, and H1 tests a partially identifiable model. This independence was chosen to stress the queue across a fixed demand range; its consequence was recognised only after H1 and motivated the post-hoc demand-coupled grid. An earlier grid that fixed incident type and lanes blocked gave $r \approx 0.90$ and was replaced because it could not vary them.

3.2.4 Stakeholders and instruments

- **Retired DIG, 27 August 2026.** A semi-structured interview on current incident handling and what drives disruption.
- **Senior DMT official, 5 October 2026.** An official with more than 20 years' experience in traffic and road work, who saw a live demonstration and completed the feedback form.

The form (Appendix A) contained the ten System Usability Scale (SUS) items [45], 1 to 5 Likert items [46] on factor relevance, usability and reliability, clearance-time estimates, open questions, and the pre-registered ranking of ten incidents: one per decile of the deployed score, shuffled so that scenario numbers do not reveal the model's order. The protocol fixed the acceptance criteria on 14 August 2026, before

data collection: a pooled Spearman correlation of at least 0.60 between expert and model ranks across the planned five to eight respondents, or a median relevance of at least 4 out of 5 for lanes blocked (capacity loss factor, CLF) and incident type (incident severity factor, ISF) with a written critique of the other factors. The DMT session was the only completed response before submission.

3.3 Experimental Procedure

1. **Scoring and hotspots.** The 500 incidents were scored with the deployed model. KDE drew the density surface and DBSCAN used haversine distance, a 0.5 km radius and four-point minimum. Moran's I used 999 permutations; Getis-Ord Gi* used eight-neighbour weights with false-discovery-rate and Bonferroni corrections. Cluster centres were compared with National Transport Commission blackspots and 10,000 random centre sets; Fisher's exact test compared membership of high-impact and other incidents.
2. **SUMO validation and sensitivity.** Each corridor scenario ran with and without the incident on identical demand. Speed reduction, edge VHL and peak-density queue were compared with the score. One-factor ablations were run, and Sequential Least Squares Programming (SLSQP) fitted bounded weights summing to one [47], checked by leave-one-road-type-out, five-fold cross-validation [48] and a fixed-weight bootstrap [49].
3. **Rerouting.** An analytical experiment compared point-queue delay with shortest-path detour time for trips crossing 78 selected incident edges. A second experiment used 96 matched scenarios on a real Galle Road–Marine Drive sub-network (159 nodes, 269 edges), with demand of 600–1,400 veh/h and one or two blocked lanes. The rerouter arm imposed full diversion around the incident edge, while the comparison arm left unblocked lanes open; network total travel time was measured for every vehicle.
4. **Dispatch (H2).** Each of 30 paired seeds streamed all 500 incidents in random order through a Poisson process at twice the fleet's estimated clearance capacity. Identical providers started from k-means depots at incident-demand centres, travelled at OSM free-flow time and served each job for its incident type's mean clearance time. Nearest-first, impact-first and an oracle ordering based on the

evaluation cost were compared at 10, 15 and 25 providers. The cost came from a gradient-boosting model of SUMO VHL using physical inputs, never the score.

5. **Transfer and stakeholders.** The score was tested on the real sub-network and against Google Routes travel times at 17 blackspots. The DMT session combined a live demonstration, discussion and unaided completion of the pre-registered form.
6. **Post-hoc checks.** The 120 scenarios were rerun with demand from the model's own profile to test internal consistency. A dispatch load sweep repeated the comparison at realised nearest-first utilisations of 0.58, 0.73 and 0.88 and added mixed and production-cost policies. These analyses were exploratory and added after the principal H1 and H2 results.

3.4 Measures and Evaluation Metrics

The main agreement measure is Pearson's r [50] between the score and SUMO speed reduction, with Spearman's ρ [51] because the score is used to rank; each factor is also correlated on its own. For the mean absolute error (MAE), speed reduction R (%) is mapped to a 1 to 10 severity by Equation (3.1). MAE is computed as scored and after a least-squares calibration of severity on the score, in points of the 1 to 10 scale, as is the 1.5-point target; both figures depend on this linear mapping.

$$S_{sim} = 1 + 9 \times (R / 100) \quad (3.1)$$

Queue length and VHL are compared with SUMO by correlation and by the median predicted-to-simulated ratio. Hotspots are measured by the haversine silhouette coefficient [27] on the clustered points, and with noise points treated as singletons, and by coverage of CRITICAL and HIGH incidents; dispatch by the paired change in total VHL against nearest-first; latency by the 95th percentile (p95) of in-process scoring time. SUS is computed from the ten item ratings s, with the odd-numbered items (1, 3, 5, 7, 9) positively and the even-numbered items negatively worded [45], and read against the average of 68 and the bands of Bangor et al. [52]:

$$SUS = 2.5 \times [\, \Sigma_{odd} (s_i - 1) + \Sigma_{even} (5 - s_i) \,] \quad (3.2)$$

Table 3.1 lists each metric with its target. Targets marked "added" were set after the proposal and were not pre-registered.

Table 3.1: Evaluation metrics and targets

Metric	Target (source)
Pearson r, score vs SUMO	≥ 0.75 (proposal, H1)
Spearman ρ, score vs SUMO	None set
MAE vs simulated severity	< 1.5 points (proposal)
Queue length error	< 20% (proposal)
VHL error	< 25% (proposal)
Silhouette (haversine)	> 0.3 (proposal)
High-impact coverage	> 80% (proposal)
Moran's I, Gi*	$p < 0.05$ (added; RQ2)
VHL reduction vs nearest-first	≥ 20% (proposal, H2)
Rerouting benefit	None set
Scoring latency p95	< 500 ms (proposal)
SUS	> 68 (added; published average)
Factor relevance	Median ≥ 4 for lanes and type (protocol)
Spearman ρ, expert vs model	≥ 0.60 (protocol)

3.5 Data Analysis

Analysis used Python with pandas [53], SciPy [54], scikit-learn [55] and NetworkX [56]; one metrics report recomputes every headline number in Chapter 6 from the stored results (Appendix C). For the expert ranking, which contained tied ranks and one unranked incident, Spearman ρ uses average ranks over the nine ranked incidents, with Kendall's τ [57] as a check. The dispatch comparison uses a bias-corrected and accelerated (BCa) bootstrap interval over the 30 paired seeds, with a Wilcoxon signed-rank test as a check. The seeds vary only the order and timing of arrivals, so these intervals reflect simulation variability, conditional on the fixed incident set, depots and cost model.

3.6 Ethics, Privacy, Security and Responsible AI

Consent and identification. The two adult professionals participated voluntarily, gave low-risk opinions about a prototype in their field and are reported anonymously. On 5 October 2026, the DMT respondent ticked the consent box and signed, dated and stamped the anonymous form; the completed form is held privately. No personal or sensitive research data were collected and no formal ethics-committee review was undertaken.

Location privacy. The score needs no driver identity. The geo-intelligence service uses the Global Positioning System (GPS) position only to identify the road, stores no request, and answers only callers with a valid token (Section 4.5). Incident history is held by the dispatch component under Sri Lanka's Personal Data Protection Act [58].

Every response exposes the five factors, supporting transparent decisions [33]. Generative artificial-intelligence use is disclosed in Appendix B.

3.7 Reproducibility and Research Artefacts

The deployed service, its OpenAPI contract and its tests are in the public Kaduna.lk platform repository [4]. The research pipeline is in the author's research repository, available to examiners on request; Appendix C lists each artefact, its purpose and output, and pins every package version, including SUMO 1.26 (Python bindings 1.27), in the research repository's requirements file. Random seeds are fixed throughout, and the service's health route returns the live weights.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The functional requirements were to calculate an explained 1–10 score and priority band; estimate congestion; expose scores, uncertainty, timelines, hotspots and statistics through a Representational State Transfer (REST) API; resolve road attributes from Global Positioning System (GPS) coordinates; and provide map, filtering, what-if, sensitive-location and exploratory placement views. These functions were delivered, although hotspots and placement remain demonstrations on synthetic records and absolute congestion magnitudes are not calibrated.

The non-functional targets covered response time, dashboard performance, security, reliability, scalability and maintainability. The service is authenticated, stateless and separated from the research pipeline; it rendered 515 incidents and scored in 9.7 ms at the in-process 95th percentile. However, one end-to-end observation took 1.45 s against the 500 ms target, dashboard load time was not measured, and the reliability target of $r \geq 0.75$ was missed at $r = 0.60$.

4.2 Component Architecture within the Complete Solution

The platform repository [4] describes the platform with the C4 (Context, Containers, Components and Code) model [59], but that model predates road resolution and the overlay and still lists the fitted weights, so Figure 4.1 was drawn for this thesis.

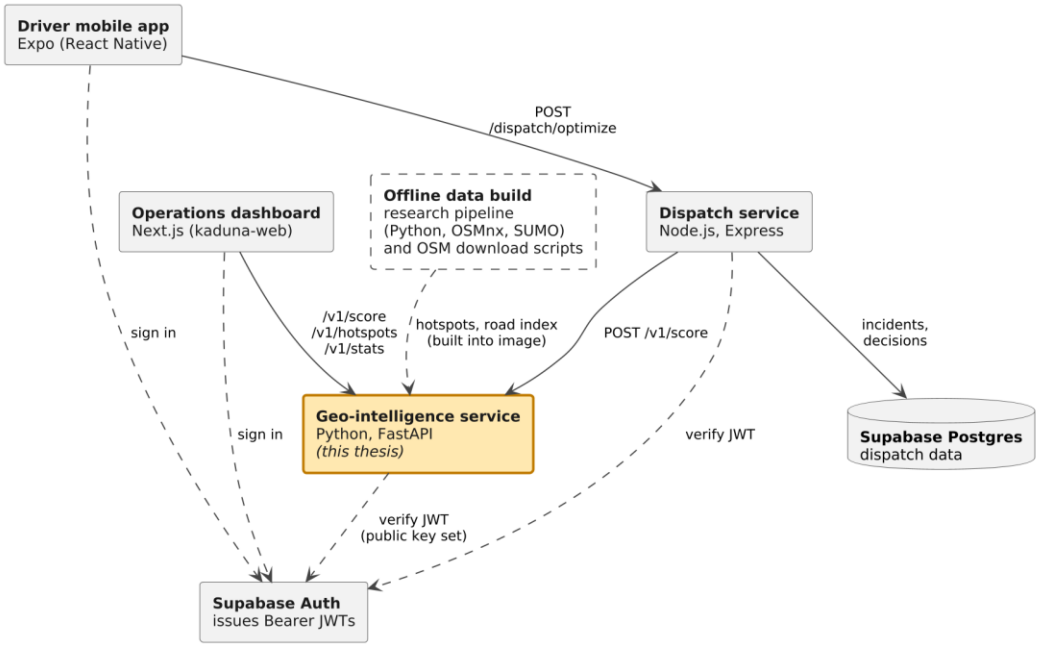


Figure 4.1: Context of the Geo-Intelligence service in the Kaduna.lk platform. Solid arrows are service calls; dashed arrows are sign-in, token verification and the offline data build (original in colour)

The stateless FastAPI service sits between the Expo mobile/dispatch path and the Next.js operations dashboard. Hotspots, the Colombo road index, points of interest and holiday calendars are built into its image; Supabase supplies authentication while dispatch owns incident records.

The component's main connection to the platform is the emergency path, called the spine:

1. The driver taps "Get help" in the mobile app, which creates an incident with the GPS position in the dispatch service.
2. The app submits the triage questionnaire, dispatch estimates the probable service type, and the app requests a dispatch optimisation.
3. Dispatch calls the scoring route with the coordinates, incident type, local hour, weekday and lanes blocked derived from triage, waiting at most 2 s before using a neutral score of 5.
4. The geo-intelligence service resolves the road, scores the incident and returns the score with its factors.
5. Dispatch combines the score with its own cost model and selects a provider.

The operations dashboard is the second consumer, requesting hotspots, statistics and scores with the user's token. The direction of these calls is the component boundary: the geo-intelligence service answers "what is happening on the road", dispatch answers "who should be sent", and the geo-intelligence service never calls dispatch.

4.3 Design Alternatives and Decisions

Weighted score instead of machine learning. A supervised model needs labelled incidents, which could not be obtained (Section 3.1). On the 120 SUMO scenarios, random forest and gradient boosting reached in-sample r above 0.99, but 120 scenarios are too few to support a learned model. A weighted sum of named factors can be argued from traffic-flow theory, shows the operator why an incident scored as it did, and can be tested against simulation without being fitted to it [33].

Spatial and simulation design. DBSCAN was preferred to k-means because it finds the cluster count, leaves isolated incidents as noise and supports haversine distance; KDE supplies the heat map [8], [9]. Synthetic corridors completed where the full city SUMO graph did not, while a real sub-network added a topology-transfer and rerouting test.

Road class from GPS. The retired Deputy Inspector General of Police (ex-DIG) stressed road importance, prompting a check that found every live incident reaching the model as a two-lane primary road. Road resolution moved into the geo-intelligence service, a direct case of stakeholder feedback improving the product.

Original weights kept in production. The deployed values are uncalibrated theory-based priors. Simulation fitting reached r = 0.93 in-sample and 0.92 held out, but against a grid unable to identify volume, time or location; it pushed those factors to their lower bounds and would move 37.2% of incidents between bands. One expert instead rated road type above traffic volume. Neither source justified replacing the priors.

Score used for triage and placement. Impact-first queue ordering was worse than nearest-first (Section 6.1.7). The deployed score therefore supplies an operator-facing band and factors, a bounded provider-cost term and exploratory placement rankings; it does not order the dispatch queue.

Sensitive locations as an overlay. Hospitals, schools, bridges and long weekends, raised by the ex-DIG, form a capped multiplier rather than a sixth factor, so the five-factor score remains the model evaluated in simulation.

4.4 Detailed Design

4.4.1 Impact score and parameter provenance

The impact score IS is a weighted sum of five factors, each normalised to 0 to 1, multiplied by 10, rounded to one decimal place and limited to 1 to 10:

$$IS = 10 \times (0.25 \cdot CLF + 0.25 \cdot TVF + 0.20 \cdot TF + 0.15 \cdot LF + 0.15 \cdot ISF) \quad (4.1)$$

The weights sum to 1, so each is the factor's maximum share of the score. Table 4.1 defines the factors. TF carries time alone, while TVF multiplies TF by a road-class peak ratio, so the two are collinear and time of day effectively carries about 0.45 of the weight. The peak ratios rank motorways highest, although Colombo's tolled expressways likely run below capacity at peak while its arterials are saturated.

Table 4.1: Definition of the five factors

Factor	Definition	Values
Capacity loss (CLF)	Lanes blocked divided by total lanes, at most 1	0 to 1
Traffic volume (TVF)	Peak volume-to-capacity ratio of the road class × hourly multiplier × day multiplier	Peak ratio: motorway 0.95, trunk 0.90, primary 0.85, secondary 0.70, tertiary 0.55, unclassified 0.50, residential 0.40, living street 0.30; other 0.60
Temporal (TF)	Hourly multiplier × day multiplier	Peaks 1.0 at 08:00 and 18:00, down to 0.02 at night; weekdays 1.0, Saturday 0.6, Sunday 0.4
Location (LF)	Road-class importance	Motorway 1.0, trunk 0.85, primary 0.70, secondary 0.50, tertiary 0.35, unclassified 0.20, residential 0.15, living street 0.10; other 0.20
Incident severity (ISF)	Ordinal severity of the incident type	Table 4.2; other 0.5

Source: peak ratios from Highway Capacity Manual (HCM) ranges [15]; other values are the author's assignment.

The score maps to four priority bands: LOW below 3, MEDIUM 3 to below 5, HIGH 5 to below 8, and CRITICAL 8 to 10; the cut-points are operational policy.

Weights. The weights are structured priors set by the author, as in a multi-attribute value model [31], rounded to avoid false precision. Delay arises when demand exceeds the capacity left after an incident (Section 2.2.1); capacity loss and traffic volume are the two terms of that condition and share half the weight, time of day changes how binding it is (0.20), and road importance and incident type are contextual modifiers (0.15 each). No formal elicitation such as the Analytic Hierarchy Process [32] was used, and the weight values were not put to an expert: the Department of Motor Traffic (DMT) form tested factor relevance, not magnitude. The weights were held fixed and graded against simulation and expert judgement, not fitted to them.

Capacities and demand. Road capacities by class (2,200 vehicles per hour for a motorway down to 300 for a residential street) and the peak volume-to-capacity ratios follow HCM ranges [15]; Section 4.4.2 states how the capacities are applied. CLF takes the lane share of capacity, but the HCM tabulates a larger loss: with one of two, three or four lanes blocked, 0.35, 0.49 and 0.58 of capacity remain, against lane shares of 0.50, 0.67 and 0.75 [15], [17]. CLF therefore understates capacity loss on multi-lane roads. The hourly curve is a double-peaked commuter profile.

Severity and clearance time. ISF is an ordinal judgement informed by clearance time (Table 4.2). Engine failure is raised to 0.7 because a disabled vehicle waits for a tow; major accidents take the maximum, as debris, investigation and emergency vehicles also block. Clearance times run from 20 minutes (out of fuel) to 120 minutes (major accident), consistent with the Traffic Incident Management (TIM) Handbook [16] and incident-duration studies [22], [23]. The DMT official's estimates differ by a mean absolute 12.1 minutes (median 10) over the seven types, with the model's values visible on the form (Section 6.3.2); they are recorded for calibration, and the model values were not changed.

Table 4.2: Incident types: generator share, lanes blocked, clearance time and severity, with the DMT official's clearance estimate

Incident type	Share of synthetic incidents	Lanes blocked	Clearance, model (min)	ISF	Clearance, DMT estimate (min)
Flat tyre	25%	1	30	0.3	30
Engine failure	20%	1	60	0.7	30
Minor accident	20%	1	45	0.5	45
Out of fuel	10%	1	20	0.2	30
Dead battery	10%	1	25	0.3	15
Major accident	8%	2	120	1.0	150
Overheating	7%	1	40	0.5	45

Source: model values from [16], [22], [23]; DMT estimates from the feedback form (Appendix A).

4.4.2 Congestion estimate

The queue, VHL and recovery estimates use a deterministic input-output queue [18], [60]. Let C be the road-class capacity (veh/h); q = C × hourly multiplier × day multiplier the demand (veh/h); μ = C × (1 − lanes blocked / total lanes) the remaining capacity (veh/h); and T the clearance time (min). The demand omits the road-class peak ratio used by TVF, so every class reaches demand equal to capacity at peak. If q ≤ μ, no queue forms; otherwise the queue length in kilometres is:

$$Q = ((q - \mu) \cdot T)/(60 \cdot K) \quad (4.2)$$

where K = 120 vehicles per kilometre is the jam density and Q is capped at 15 km. The (q − μ) · T / 60 excess vehicles each wait T/4 on average, which gives VHL; this fixed mean wait is below the cumulative-count delay of Section 2.2.1 by a factor of at least two. If q < C, the queue drains at the spare capacity, so recovery takes 60 · Q · K / (C − q) minutes; otherwise it is set to T/2 by convention. Recovery time is capped at 180 minutes.

Two conventions set the estimator's magnitudes. The class capacities have HCM per-lane magnitudes but are used as whole-road capacities, and the per-lane jam density of 120 veh/km is applied to the whole road. Demand in the SUMO grid, set as multiples of these capacities, therefore stays well below the simulated corridor's physical capacity and rarely binds, a likely root cause of the queue and VHL over-

prediction of Section 6.1.4; and CLF understates the HCM incident-capacity reductions (Section 4.4.1). A sensitivity analysis of these conventions leaves the deployed model unchanged (Sections 6.1.4 and 7.5).

4.4.3 Uncertainty, timeline and sensitive-location overlay

The uncertainty band re-scores the incident 1,000 times, drawing clearance time from a log-normal distribution (the type's mean, 30% coefficient of variation), scaling ISF by drawn over mean clearance time (at most 1), and moving lanes blocked by −1, 0 or +1 within 1 and the lane count. It returns the mean and the 5th and 95th percentiles. The timeline is triangular: relative impact rises linearly to 1 at clearance and falls linearly to 0 over the recovery time, and stays at 0 if no queue forms.

The sensitive-location overlay multiplies the base score by 1.0 to 1.35: up to 15% near a hospital (decaying over 400 m, ignored beyond 1.2 km); 10% within 150 m of a school during weekday arrival and dismissal hours; 10% within 100 m of a bridge, which offers no lateral detour; 5% within 200 m of a market at busy hours; and 10% on a working day before three or more non-working days. These priors are capped at 35% in total. The overlay is a priority adjustment for the consequence of blocking access near such sites, not an estimate of traffic delay. The adjusted score (base score times the factor, capped at 10) is reported beside the base score, which dispatch uses.

4.4.4 API and data model

The versioned API exposes health, score, uncertainty, timeline, hotspot and statistics routes; all except health require a token. A score request carries coordinates, incident type, lanes blocked and local time, optionally with road attributes and date. The response adds the band, factors, relative congestion estimates, matched-road provenance and sensitive-location overlay. Hotspot responses include centroid, size, mean score, dominant road and incident type, peak hour and composite risk.

4.5 Standards and Professional Practice

Interfaces. The REST interface [61] uses JSON, HTTP status codes and an OpenAPI 3.1 contract [62] checked against the registered routes.

Security. Supabase Bearer JSON Web Tokens are verified for signature, issuer, audience and expiry [63]. The service fails closed when authentication is

unconfigured, rate-limits users and restricts browser origins, addressing broken authentication risks in the OWASP API Security Top 10 [64]. Role restriction for analytics routes remains disabled in production (Section 7.5).

Engineering practice. Requirements follow ISO/IEC/IEEE 29148 [65], contracts and tests run in continuous integration, and personal data is governed by Sri Lanka's Personal Data Protection Act [58].

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The private research repository builds the incident set, runs Simulation of Urban MObility (SUMO) experiments and produces Chapter 6's evidence. The public platform repository [4] contains the production FastAPI/Pydantic service, Next.js/Leaflet operations views, OpenAPI contract, Docker image and tests. Analysis uses SciPy, scikit-learn, OSMnx and NetworkX; Supabase supplies token authentication. Static datasets are built into the stateless service image.

5.2 Component Implementation

5.2.1 Research pipeline

The pipeline writes comma-separated-value or GeoJSON outputs between network download, incident generation, scoring, hotspot analysis, SUMO validation and sensitivity stages. Follow-up analyses read these outputs without changing the deployed model, and one report recomputes every headline metric (Appendix C). SUMO inputs are generated programmatically and outputs parsed from XML; the real sub-network preserves OSM edge lengths and lanes, while variable speed signs implement blockage.

5.2.2 Scoring service

Routes and schemas, scoring, road resolution, the sensitive-location overlay and token verification are separated by concern. Production uses the original weights, and five golden incidents pin its output to the research model at release. Numeric fields are range-checked; missing hotspot data returns service-unavailable, while optional road, point-of-interest and holiday data degrade to documented defaults.

5.2.3 Road resolution

The service resolves each coordinate against 34,265 Colombo OSM ways by bounding-box filtering and point-to-segment distance. A match within 50 m supplies road class and lane count; otherwise the documented two-lane-primary fallback applies. Only 7% of ways have a lanes tag, so most counts use class defaults, and

OSM commonly records both directions together. Provenance and match distance are returned with the score.

5.2.4 Offline data products

Offline builds produce the hotspot file, road index, 147 hospitals, 613 schools, 51 markets, 401 bridge segments and 2026–2027 holiday calendars. Rebuilding them updates the service without code changes.

5.2.5 Operations dashboard view

The dashboard lists live scored incidents and factor breakdowns, draws hotspot, heat-map and blackspot layers, and provides a what-if panel. A generated 312-case parity suite and exhaustive 254,016-input check matched the Python model. Placement recommendations rank clusters by the relative VHL reduction expected from earlier arrival, identify unit type and peak hour, and remain exploratory because absolute congestion magnitudes are uncalibrated. The product gives no detour advice.

5.3 Integration with the Other Components

The author implemented dispatch's geo client and corrected its traffic term to scale with each provider's travel time. The client maps triage output to incident type and lanes blocked, sends Colombo local time, forwards the access token and uses a neutral score of 5 after a two-second timeout. Decisions log score provenance and latency. OpenAPI contracts govern the interface, continuous integration checks them, and a post-deployment smoke test exercises the complete path.

5.4 Verification and Software Testing

All 146 geo-service tests pass: 31 API, 46 authentication/authorisation, 29 overlay, 19 road-resolution, 10 rate-limit/origin, six scoring and five OpenAPI tests. Continuous integration also checks data parity, dispatch against PostgreSQL and the web tests. A code-versus-thesis audit exposed and corrected CI, parity, placement, provenance, rate-limit and holiday-calendar defects before version 0.2.0. Latency was measured over 1,000 warmed in-process calls of a worst-case road scan; one authenticated Sri Lanka-to-Europe call measured network-dominated end-to-end performance.

5.5 Deployment and Operation

Dokploy runs the Dockerised service beside the other components on a European virtual private server behind HTTPS. The public health route exposes version, active weights and dataset state; configuration uses environment variables and public authentication keys. Back-end releases are manual, followed by health, authenticated-score and dispatch checks. The stateless service supports rollback by redeployment and horizontal replication.

6 RESULTS AND VALIDATION

6.1 Research Results

This chapter reports results against the targets in Table 3.1. The model was evaluated on 500 synthetic incidents placed on the real Colombo OpenStreetMap (OSM) road network, and against the Simulation of Urban MObility (SUMO) microsimulator [35] on a 120-scenario grid of synthetic corridors. All numbers were recomputed by the validation scripts in Appendix C; correlations that decide a verdict carry 95% confidence intervals (CIs).

6.1.1 Impact score against simulated congestion (H1)

Simulated speed reduction ranges from 0% to 99.6% (mean 19.3%). The deployed score (Equation 4.1) correlates with it at Pearson $r = 0.60$ (95% CI 0.47–0.70) and Spearman $\rho = 0.67$ (0.55–0.77) [51] (Figure 6.1). The CI excludes the H1 target of $r \geq 0.75$, so H1 is not supported. The blockage pattern explains 99% of the variance in speed reduction, so both coefficients mainly reflect how well the score separates the 25 two-lane blockages: among single-lane scenarios $r = 0.47$, and with the scenarios reweighted to the generator's incident mix $r = 0.49$. As Section 6.1.2 shows, the grid varies only lanes blocked in a way the simulator responds to, so H1 tested a partially identifiable model; the miss stands, and the demand-coupled grid (Section 6.1.3) is supplementary.

Over the 500 incidents the mean score is 5.54 (median 5.60), with 18 CRITICAL (3.6%), 300 HIGH (60.0%), 171 MEDIUM (34.2%) and 11 LOW (2.2%), so the band alone separates incidents coarsely.

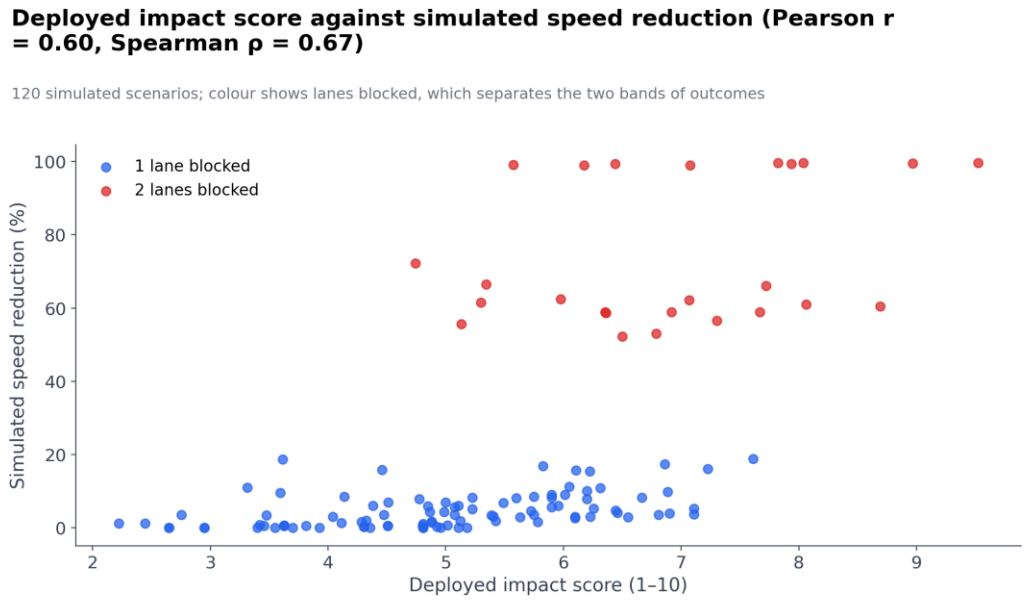


Figure 6.1: Deployed impact score against SUMO speed reduction on the 120-scenario grid (r = 0.60, ρ = 0.67), coloured by lanes blocked (original in colour)

6.1.2 Factor contribution and sensitivity

Lanes blocked (capacity loss factor, CLF) correlates with speed reduction at r = 0.88 and incident type (ISF) at r = 0.80, but the latter inherits the two-lane blockage assigned only to major accidents; among single-lane scenarios its r is −0.07. TVF, TF and LF each have $|r| \leq 0.09$ because their inputs were sampled independently of demand. Dropping TVF, TF or LF raises the full score's r to 0.78, 0.72 or 0.64, while dropping CLF lowers it to 0.35.

Simulation-fitted weights (CLF 0.500, TVF 0.050, TF 0.050, LF 0.071, ISF 0.329) reach r = 0.93 in-sample and 0.92 under leave-one-road-type-out and five-fold cross-validation. They fit blockage in this grid rather than independently validating all factors and remain sensitivity-only.

6.1.3 Demand-coupled simulation (supplementary)

In the post-hoc grid driven by the model's own demand rule, score agreement with speed reduction falls to ρ = 0.35 because slowdown remains blockage-driven. Against total VHL, however, the score reaches ρ = 0.91 (95% CI 0.87–0.93), TVF 0.66 and TF 0.57; a simple volume × duration × capacity-loss proxy reaches 0.92. This demonstrates internal consistency with the demand rule, not external validity.

6.1.4 Error magnitude and congestion estimates

With speed reduction R mapped to a 1–10 severity (Equation 3.1), the mean absolute error (MAE) is 3.16 points as scored and 1.68 after a least-squares calibration of severity on score, both above the 1.5 target and both dependent on the linear mapping of Equation 3.1.

The point-queue congestion estimator [18], [60] is separate from the score. Its VHL and queue outputs rank SUMO's at Spearman $\rho = 0.71$ and 0.61 (Pearson $r = 0.80$ and 0.71). It predicts no queue in 34 of the 120 scenarios and over-predicts by a median of 20 times for VHL and 12.4 times for queue length (35 and 17 times where both are non-zero), missing both magnitude targets; its outputs are a relative index only.

A capacity sensitivity check reduced but did not remove the magnitude error and often predicted no queue. The Highway Capacity Manual incident factor tracked slowdown slightly better than lane share alone ($r = 0.90$ against 0.88) but lowered the full score's r to 0.57. The deployed estimator is therefore treated only as a relative index.

6.1.5 Hotspot analysis

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [8] (haversine metric, 0.5 km radius, four points minimum) finds 25 clusters, with 356 incidents (71.2%) as noise; the kernel density estimation surface supplies the heat-map layer (Figure 6.2). The haversine silhouette coefficient [27] is 0.64 on the 144 clustered points (target above 0.3) and 0.08 with noise points treated as singletons, so it measures the separation of the dense clusters only.

Of 318 CRITICAL and HIGH incidents, 110 (34.6%) lie in a hotspot (target 80%), against 34 of the 182 others (18.7%): an odds ratio of 2.3 (Fisher's exact $p = 0.0001$). The noise share and the coverage miss are two views of one parameter choice: at 0.5 km and four points, most incidents on long arterials are not dense enough to cluster. Ten of the 25 cluster centres lie within 1.5 km of one of eight documented accident-prone areas, against 5.0 expected for centres placed at 25 random incident locations ($P(\geq 10) = 0.02$); 13 lie within 1.5 km of any of the 18 sites on the National

Transport Commission (NTC) 2024 accident-prone list, against 7.3 expected (P = 0.01).

Moran's I [28], on incident counts in the 173 non-empty cells of a 20 × 20 grid, is 0.22 ($p \leq 0.001$, 999 permutations), describing the clustering the generator builds in. Getis-Ord Gi* [29], on the impact score with the same neighbourhood, flags 40 of the 500 locations at uncorrected $p < 0.05$; none remains significant after false-discovery-rate or Bonferroni correction, so no local hotspots are claimed.

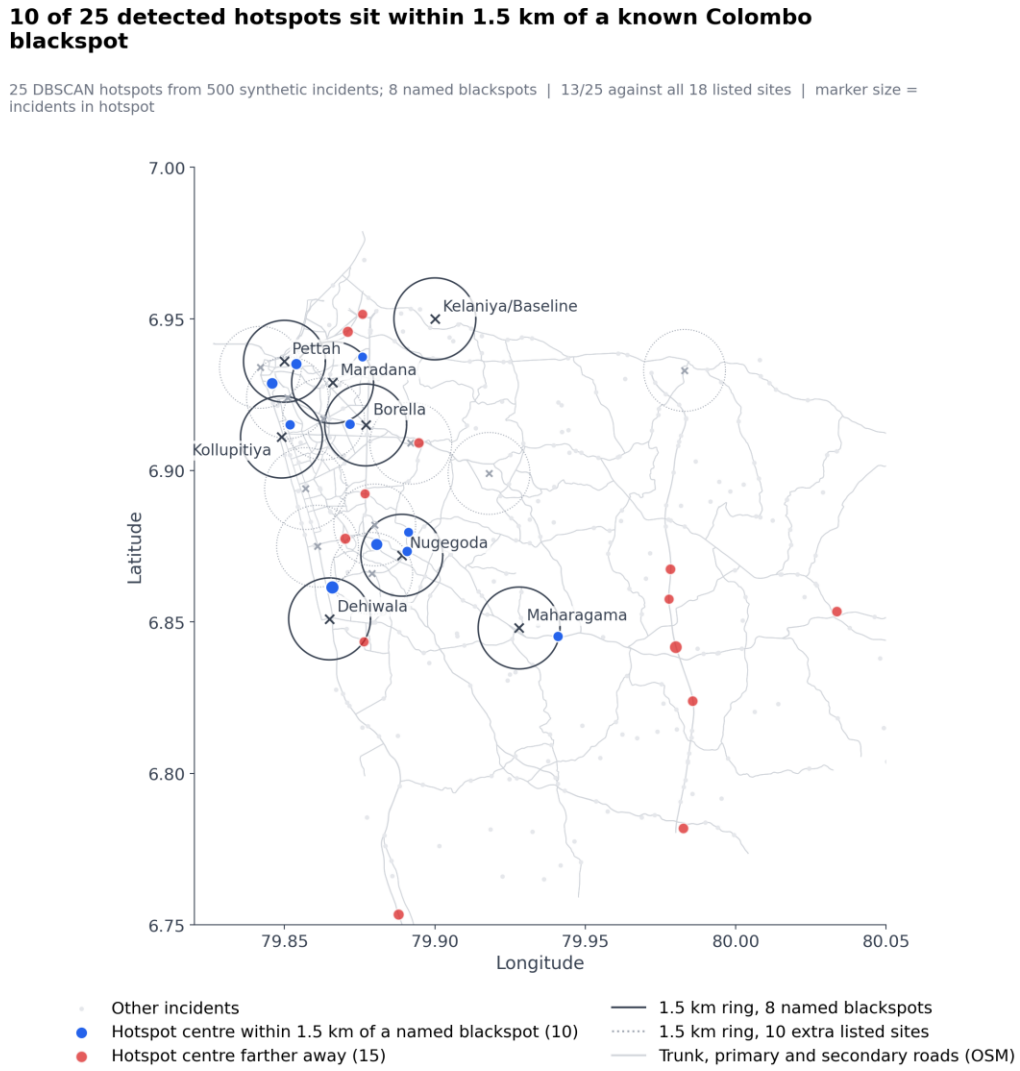


Figure 6.2: DBSCAN hotspot centres on the 500 synthetic incidents, with 1.5 km rings around the eight named blackspot areas (solid) and the ten further listed sites (dotted) (original in colour)

6.1.6 Rerouting around a blocked segment

Microsimulation on a real sub-network. SUMO was run on a real Colombo sub-network (159 nodes, 269 edges) in which the incident blocks Galle Road and Marine Drive is the parallel route. Each of 96 scenarios ran once with and once without a rerouter on identical demand (Table 6.1). The comparison is not like for like: with the rerouter on, the whole incident edge was closed and every approaching vehicle diverted (full compliance); with it off, only the blocked lanes were slowed and two or three lanes stayed open.

Table 6.1: Rerouter off against on, real Colombo sub-network (96 matched pairs; vehicle-hours per scenario)

Metric	OFF	ON	Change (95% CI)
Network total travel time, 40-minute window	18.0	36.9	+18.8 (16.8 to 20.8)
Network time loss, to arrival	1.4	9.1	+7.7 (6.2 to 9.3)
Galle Road corridor travel time	16.2	14.9	−1.2 (−1.4 to −1.0)

The rerouter roughly doubled network travel time, by 18.8 vehicle-hours per scenario (20.9 when counted to arrival), and the rise grew with demand: +8.5, +15.6 and +32.4 vehicle-hours at 600, 1,000 and 1,400 vehicles per hour. About 530 vehicles per run left the corridor on routes about 686 m longer, so corridor travel time fell while time loss per remaining corridor vehicle rose. The +0.87 reported in earlier drafts as network delay was this corridor time loss, which excludes all detour travel. Forced, full diversion around a partially blocked multi-lane arterial was counter-productive in this network.

Analytical estimate on the Colombo graph. Of 78 sampled incidents, 62 produced trips crossing the blocked edge and 16 produced none. Across those 62 incidents, 437 of 439 trips had a viable detour; one incident had no viable detour for any sampled trip. For the 61 incidents with detours, the median estimated saving of 13.3 minutes per trip is an uncalibrated model output: its queueing delay comes from the estimator of Section 6.1.4, and corrected for that estimator's 12- to 20-fold over-prediction the median falls below 0.5 minute, with 53–64% of trips gaining. It therefore does not establish a rerouting benefit.

6.1.7 Impact-ordered dispatch (H2)

H2 was tested by closed-loop dispatch simulation on the Colombo network (30 paired seeds per fleet size). Each incident's cost is its incident-edge VHL as predicted by a gradient-boosting model fitted on physical features (road-class capacity, lanes blocked, hour-of-day demand, excess demand, clearance duration), never the score or its weights; its held-out (leave-one-road-type-out) Spearman ρ against SUMO VHL is 0.89, and its rank correlation with the score 0.51. H2 is rejected (Figure 6.3): impact-first queue ordering is 23.7%, 38.2% and 64.7% worse than nearest-first at 10, 15 and 25 providers. Ordering by the cost model's own predicted delay (oracle ordering) tied with nearest-first at 10 providers (−0.5%, 95% CI −1.4 to +0.3; 12 of 30 seeds won) and was worse at 15 and 25 (−6.3% and −23.3%), which bounds only the pure queue orderings tested. All percentages in the text are paired means over seeds.

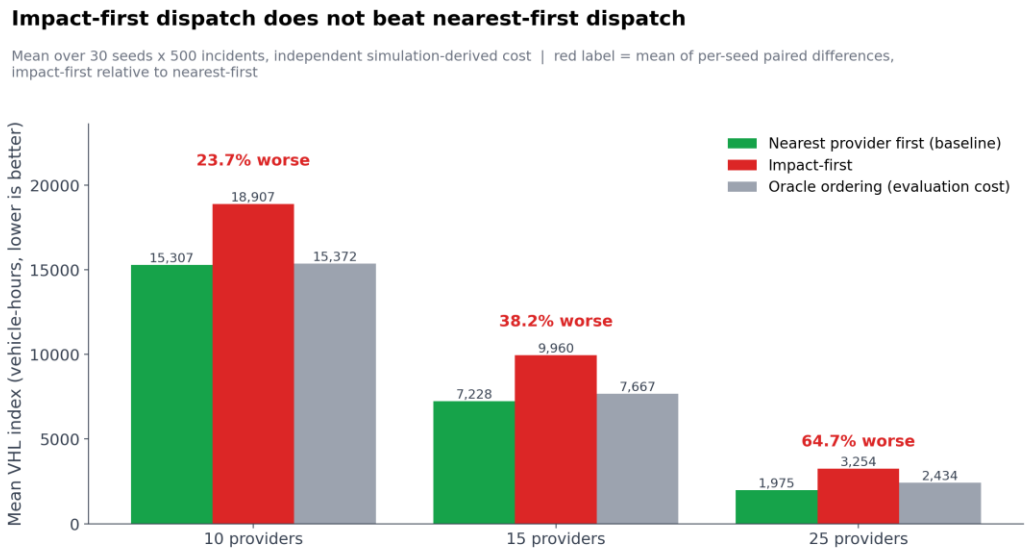


Figure 6.3: Mean total VHL (vehicle-hours) by dispatch policy and fleet size (30 paired seeds; lower is better). Percentage labels are means of the per-seed paired differences (23.7%, 38.2%, 64.7%); the ratio of the plotted means gives 23.5%, 37.8% and 64.8% (original in colour)

The mechanism is travel distance: under overload any priority order sends providers farther. At 15 providers mean response rises from 297 minutes (nearest-first) to 600 (oracle), while deferred incidents accumulate delay.

Realistic fleet load (supplementary). At realised utilisations of 0.58, 0.73 and 0.88, impact-first was 0.2%, 1.3% and 19.1% worse. At 0.88 its mean response was 41.2 minutes against 20.1 and it won none of 30 seeds. A bounded mixed policy stayed within ±0.4% of nearest-first, while ordering by the production provider cost lost 1.2% and 9.9% at the two higher loads. H2 remains rejected.

6.1.8 Transfer to real topology and live traffic

On the real sub-network, the deployed score correlates with speed reduction at r = 0.20 in the 96 runs without rerouting and r = 0.07 across all 192, while lanes blocked alone reaches r = 0.96. Lanes blocked took two levels there (mean slowdowns 2.3% and 4.4%) and demand varied independently of hour, so the test cannot rank the factors. At 17 listed blackspot sites, live peak-hour delay from Google Routes against the score for one hypothetical incident (three distinct values) gives ρ = 0.06 (95% CI −0.43 to 0.53). The anchor compares recurrent congestion with a score for incident-induced delay, a different construct, and at this size can neither support nor refute the location factor.

6.1.9 Summary against targets and baselines

Table 6.2 consolidates the principal evidence. The score, congestion magnitude, hotspot coverage, dispatch and end-to-end latency miss their proposal targets; cluster separation passes on clustered points.

Table 6.2: Principal results against targets and baselines

Evaluation	Result	Verdict
Score vs SUMO; target r ≥ 0.75	r = 0.60; lanes blocked alone 0.88	H1 not supported
Congestion magnitude	Queue 12.4× and VHL 20× too high	Relative index only
Hotspots	Silhouette 0.64; coverage 34.6% vs 80%; 10 centres near blackspots vs 5 by chance	Separation passes; coverage misses
Impact-first vs nearest-first	19.1% worse at utilisation 0.88; 23.7–64.7% worse under overload	H2 rejected
Forced rerouting	Network total travel time 18.0 → 36.9 vehicle-hours	Full diversion rejected
Service latency	In-process p95 9.7 ms; one end-to-end call 1.45 s vs 500 ms	Model fast; end-to-end target not met
Expert evaluation	Ranking ρ = −0.05; relevance arm 5/5; one respondent	Relevance criterion met indicatively; ranking inconclusive

6.2 Product and Technical Results

The stateless FastAPI service is deployed from the Kaduna.lk repository [4], requires Supabase tokens except for health, and reports the original weights. Version 0.2.0 passes 146 tests. Across 1,000 warmed worst-case calls, in-process p95 was 9.7 ms; one Sri Lanka-to-Europe end-to-end call took 1.45 s.

The interview with the retired Deputy Inspector General of Police (ex-DIG; Section 6.3.1) prompted a check that revealed every incident had been scored as if it lay on a two-lane primary road. Road class and lane count are now resolved from the incident's Global Positioning System (GPS) position against 34,265 bundled OSM road ways; the same engine-failure incident (one lane blocked, Monday 08:00) now scores 7.8 on a trunk road in Colombo Fort and 5.5 on a Nugegoda residential street.

The dashboard (Figure 6.4) shows scores, factors, map layers, a scoring log, what-if tool and model-accuracy view. Its exploratory placement view ranks clusters by relative avoided VHL and names the unit type and peak hour.

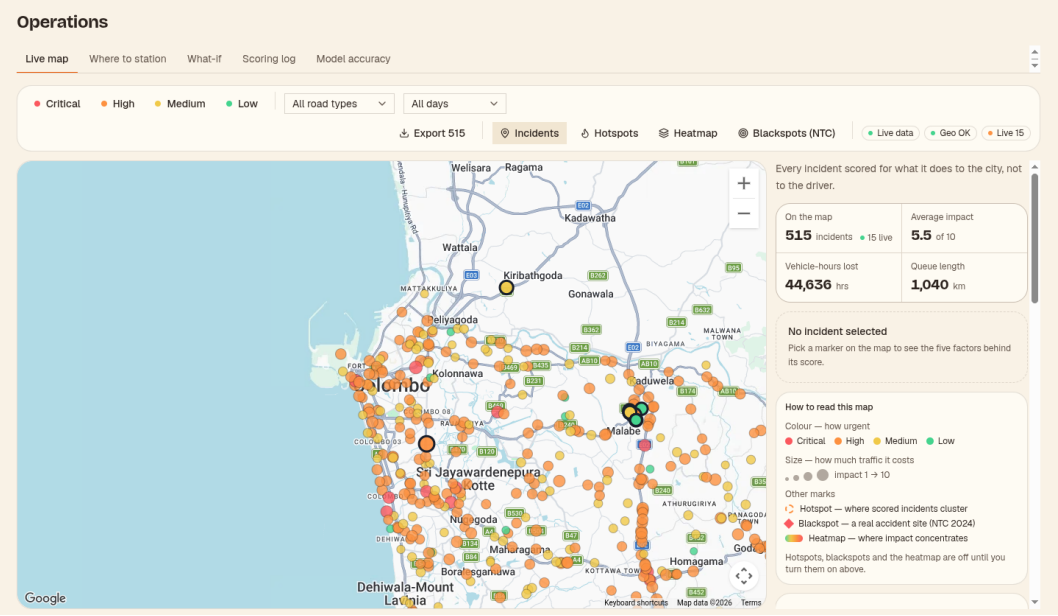


Figure 6.4: Operations dashboard, live map, showing 500 synthetic incidents plus 15 live incidents at capture, coloured by priority band. The vehicle-hours and queue-length tiles are relative index values from an uncalibrated model (original in colour)

6.3 Stakeholder Validation

The ex-DIG gave a semi-structured interview, and a senior Department of Motor Traffic (DMT) official completed the pre-registered form (Appendix A) after a live

demonstration. No operations-desk user was available; the DMT respondent represents the traffic-authority group, not the primary user.

6.3.1 Ex-DIG interview (27 August 2026)

The respondent stated that the formal incident pathway operates on expressways only and "there is no similar system to help locals" on ordinary roads. Unprompted, the ex-DIG described a weekday and direction effect (heavy traffic into Colombo on Mondays, out on Fridays, worst on long-weekend eves), which the model's uniform weekday multiplier does not represent. His emphasis on road importance prompted the check that exposed the road-class defect described in Section 6.2.

6.3.2 DMT structured evaluation (5 October 2026)

The respondent is a senior DMT official with more than 20 years' experience in traffic and road work, who chose to sign the form although it was anonymous by design.

Usability and relevance. The System Usability Scale (SUS) [45] score is 80.0 ("good" [52]); an alternative reading of a corrected item gives 70.0. Factor relevance averaged 4.4 out of 5: lanes blocked, road type and incident type scored 5; time 4; and traffic volume 3. Both usability and trust items scored 5. With one respondent after a guided demonstration, these results are indicative only.

Pre-registered scenarios and duration. Nine of ten shuffled scenarios were rated. Agreement with the model was inconclusive (Spearman $\rho = -0.05$, 95% CI −0.69 to 0.63; Kendall's $\tau = -0.03$), against a protocol designed for five to eight respondents. The alternative protocol arm—relevance of at least 4 for lanes blocked and incident type—was met with ratings of 5, but with $n = 1$ is indicative rather than validation. The largest ranking disagreements were major roads at modelled off-peak times, suggesting that road class may be under-weighted. Expert clearance estimates differed from defaults by a mean 12.1 minutes (median 10), with the largest gaps for major accidents (150 vs 120 minutes) and engine failure (30 vs 60). The respondent independently confirmed weekday and direction effects.

6.3.3 Data availability finding (DMT)

The official stated that DMT keeps no timed, location-coded incident records and receives only major incidents, while minor cases go to the Police. This single-source institutional finding explains the use of simulation and expert judgement and identifies Kaduna.lk's timestamped, geocoded records as possible future calibration data.

6.4 Integration Results

The full path was verified locally on a physical Android device and twice in production. After version 0.2.0, triage called dispatch, which logged a geo-sourced score of 9.2 rather than the neutral fallback. This demonstrates contract, token forwarding and fallback operation; it does not establish improved production decisions. The bounded term cannot reorder otherwise identical providers, and the score does not order the queue.

7 DISCUSSION

7.1 Interpretation of Findings

Meaning of the score. Moderate rank agreement and large absolute error describe a compressed ordering rather than calibrated impact. Because most incidents fall in the HIGH band, operators should inspect the numeric score and factors. Lanes blocked dominates because the simulation responds to capacity removed; incident duration matters in theory but mean speed drop cannot represent it fully [18], [22]. Demand changes the number of affected vehicles rather than the blockage-driven speed loss, explaining why the score ranks its own coupled total-delay rule better than speed reduction.

Why five factors and the original weights remain. Lanes blocked alone outperforms the full score on every external outcome measured here. Road class, time and volume remain theory-based priors for effects that the simulation could not identify, not empirically established improvements. Fitted weights suppress those factors, while the single expert placed road type above traffic volume and disagreed most on major off-peak roads. Neither compromised source justifies moving more than a third of incidents between bands; reweighting requires real records or agreement from several raters.

Impact informs triage, not queue order. Not even an ordering by the evaluation cost itself beat nearest-first, and impact-first stayed worse at realistic load (Section 6.1.7). Among the orderings tested, travel distance is therefore the binding constraint: any priority order sends providers farther, and the extra travel outweighs the delay saved, while a small bounded travel weight (the mixed policy) is neutral. The score informs triage through the band, factors and placement ranking an operator sees (Section 6.4).

Forced rerouting was counter-productive. Diverting every vehicle from a road with two or three lanes still open roughly doubled network travel time, though the corridor gained (Section 6.1.6): delay moved onto a longer, congesting detour. Rerouting advice therefore needs partial compliance and a detour-capacity check.

7.2 Comparison with Literature Baselines and Existing Solutions

Incident impact. Unlike simulation studies that directly measure impact [5], [11], this study grades a separate interpretable score. Capacity loss dominating agrees with HCM theory and recorded-data models, while those supervised alternatives require incident logs unavailable here [6], [7].

Rerouting. Smith et al. found that runtime rerouting reduced travel time [11]. Here forced, full diversion raised network travel time, as the queueing argument of Section 2.3.3 predicts when the detour is longer and has limited spare capacity.

Hotspots. Zhang et al. [10] and Zheng et al. [9] pair density clustering with spatial statistics on recorded accidents. This study follows that design but ranks clusters by impact rather than count, one answer to the disagreement Elvik [24] found among agencies on whether severity enters blackspot definitions.

Dispatch. Roadside dispatch systems [13], [14] order by service-level priority, not network impact. The H2 result agrees with the ambulance-dispatch literature: Jagtenberg et al. [39] showed that departing from closest-idle dispatch pays only for particular objectives and depends on travel times and future coverage. Here, even ordering by the evaluation cost did not offset the extra travel that priority ordering causes under load, so closest-first remains the policy to beat.

7.3 Research Contribution

1. **An interpretable incident-triage score that can be built and graded without incident logs.** It was evaluated factor by factor against microsimulation for Colombo on synthetic incidents and deployed on a Sri Lankan roadside-assistance platform; in simulation, lanes blocked carries the signal.
2. **A characterised negative result on impact-ordered dispatch, under the policies tested.** It was measured with a cost not derived from the score, is robust to realistic fleet load, and was compared with an ordering by the cost model's own predicted delay (oracle ordering). It places impact in triage rather than queue ordering.
3. **A pre-registered, multi-source validation protocol for a setting where incident records could not be obtained.** It combines simulation, cross-

validation, a transfer test to real topology and a pre-registered expert instrument, and identifies the platform as a possible future source of such records.

These contributions are supported by fixed targets, explicit misses, confidence intervals, multiple-testing corrections and a dispatch cost independent of the score. Stakeholder feedback also led to a diagnosed road-class defect, a deployed fix and a regression test.

7.4 Product Contribution

A deployed, authenticated and tested scoring service consumed by dispatch in production; road-class resolution from the incident's position prompted by stakeholder feedback; and an operations dashboard whose placement view converts hotspots into station recommendations, rated 80.0 on the System Usability Scale (SUS) by a senior Department of Motor Traffic (DMT) official, a single respondent and so indicative only.

7.5 Limitations and Threats to Validity

Data. The 500 incidents are synthetic on real geometry. The generator ties blockage to type and favours arterials, affecting hotspot and blackspot results; clustering uses straight-line distance and ignores exposure. Observed incident outcomes were unavailable.

Simulation. Slowing a full 1 km lane to 0.1 m/s makes speed reduction largely a measure of vehicles trapped in blocked lanes. The one-seed corridor runs omit junctions, spill-back and insertion delay, censor long incidents and sample demand independently of hour. Accordingly, the grid identifies blockage and its $r = 0.60$ depends on scenario mix. Real-subnetwork rerouting imposed full closure and 100% compliance, so it tests forced diversion rather than realistic advice.

Model parameters. Per-lane HCM capacities and jam density are applied as whole-road values, contributing to 12.4× queue and 20× VHL overestimation. The point queue cannot spill upstream; recovery includes a convention; lane share understates incident capacity loss; and TVF and TF duplicate time inputs. United States

parameters do not reflect Colombo's vehicle mix, OSM lane tags often combine directions, and the sensitive-location overlay is unvalidated.

Dispatch. The cost model inherits the grid's weak time effect and accrues linearly with response time, favouring nearest-first relative to Equation 2.1. Providers are identical, travel at free-flow speed and start at fixed depots; intervals vary arrival order only. The operational benefit of triage was not tested.

Stakeholders. One of five to eight planned respondents completed the instrument after a guided demonstration, and no operations-desk user participated. One usability item incorrectly mentioned detours, and displayed model durations may have anchored expert estimates.

Engineering. One end-to-end latency observation missed the target and no load test or road-matching accuracy study was run. Operations-role enforcement is disabled pending role-bearing tokens, score provenance is logged but not stored, first-request road-index loading adds about 400 ms, and golden tests do not guard the private research copy.

7.6 Implications

For traffic authorities: recording each incident's coordinates, road, start and clearance times, through the Police for minor incidents and the DMT for major ones, would create the missing ground truth. **For platform builders:** keep travel distance as the primary dispatch criterion, with impact at most a bounded cost term whose effect is tested before it is relied on; offer detours only with partial compliance and a detour-capacity check; and evaluate scores with costs not derived from them.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The component returns an explained 1–10 score, matched road and sensitive-location overlay, and turns incident history into hotspots and exploratory unit-placement rankings. Its value is supporting which incident an operator examines first and where units might wait. The 25 hotspots are a synthetic-data demonstration, congestion magnitudes are relative indices, and detour advice is excluded because forced diversion roughly doubled simulated network total travel time.

Primary users are operations staff and dispatch; prospective customers are assistance and fleet operators, insurers and road authorities. Drivers benefit indirectly, while authorities could use aggregate impact and hotspot views.

8.2 Market and Competitive Position

Table 8.1 compares the main alternatives.

Table 8.1: Capabilities of existing alternatives compared with the Geo-Intelligence component

Capability	**Google Maps and Waze**	**Insurer and private hotlines**	**Kaduna.lk Geo-Intelligence**
Incident reporting	Crowd reports	Telephone	Mobile app
Live congestion and driver rerouting	Yes	No	No
Per-incident impact score with explained factors	No	No	Yes
Priority band for operator triage	No	Manual judgement	Yes
Hotspots and unit placement recommendations	No	No	Yes
Integrated with provider dispatch	No	Manual	Yes

Navigation platforms are the strongest incumbents, but they expose no per-incident impact score for deciding what to clear first and are not connected to the businesses that clear incidents. Sri Lankan hotlines and garages appear to handle requests in arrival order. The component's position is therefore an explainable impact score on every incident of a roadside-assistance platform, for an operator who must triage rather than a driver who must navigate. Because a navigation platform could

compute a similar score, the lasting advantage is the integration with response providers and the local incident data the platform accumulates.

8.3 Business and Sustainability Model

Four routes are proposed.

1. **Business-to-government (B2G) licence.** Authorities could license the operations view: the live incident list with impact scores, the hotspot map and the placement recommendations. A realistic first step is a pilot with one authority in one area of Colombo, comparing the score with the authority's judgement of real incidents.
2. **Business-to-business (B2B) API.** Insurers, fleet operators and roadside-assistance companies could subscribe to the scoring and hotspot routes of the application programming interface (API). The service is already versioned, contract-documented, stateless and rate-limited per user, so a paid tier would mainly need per-customer keys and request metering.
3. **Data asset.** Per-incident, timed, geocoded records were not obtainable for this study (Section 6.3.3), while every incident handled through Kaduna.lk carries its coordinates, start time and clearance time. In aggregate, this record is the dataset needed to calibrate the model and is valuable to road safety planners and insurers.
4. **Open-data partnership.** Aggregated, de-identified statistics, such as counts and mean impact by road segment and hour, could be published with an authority or a university, building the trust a government licence requires.

The stateless service shares the platform host and uses open-source software and OpenStreetMap, keeping operating cost low. A pilot should test per-seat dashboard and per-request API pricing, with success measured by ranking agreement with authority judgements and completeness of coordinates, start and clearance times.

8.4 Scalability, Intellectual Property and Compliance

Scalability. OSM and the factor formula can be rebuilt for other areas, but each city needs local profiles, sensitive locations and calibration. National coverage would need a spatial index; the stateless service can replicate horizontally.

Intellectual property. The method combines well-established factors, so no patent is sought. The protectable assets are the source code (owned by its authors subject to the university's regulations on student work), the curated Sri Lankan configuration and, in time, the incident dataset. Publishing the model openly would support public-sector trust, while the integration, data and operated service remain the offering.

Licences. The software stack is permissively licensed; SUMO is used offline. OSM-derived maps and data require attribution and Open Database Licence compliance [44], [66], while optional Google Maps use invokes its platform terms.

Personal data. Location data that can be linked to a driver is personal data under Sri Lanka's Personal Data Protection Act, No. 9 of 2022 [58]. The geo-intelligence service receives only coordinates, time, incident type and lanes blocked, and stores neither identity nor requests; incident records are held by the dispatch service under the platform's consent and access controls. The data-asset and open-data routes would require explicit consent at reporting, a defined retention period, and aggregation of locations before any release.

8.5 Contribution to the Sustainable Development Goals

The component relates to three of the United Nations Sustainable Development Goals (SDGs) [67].

- **SDG 3.6:** timed incident records and quicker recognition of disruptive incidents can support road-safety management [2].
- **SDG 9.1:** impact and bridge overlays support resilient-infrastructure planning.
- **SDG 11.2:** hotspot and access-sensitive views can support more reliable urban transport.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis developed the Geo-Intelligence component of Kaduna.lk to estimate how much an individual roadside incident disrupts traffic in Colombo. The component combines an interpretable five-factor impact score, a congestion estimator and a hotspot analysis in a live, tested geo-intelligence service that the dispatch component consumes.

Against a 120-scenario SUMO grid, the score ranks simulated incidents moderately (Spearman $\rho = 0.67$; Pearson $r = 0.60$, short of the 0.75 target). The grid identifies lanes blocked ($r = 0.88$) rather than all five factors, and transfer to a real sub-network is weak ($r = 0.20$). The congestion estimator ranks VHL and queue length but over-predicts their magnitudes by 20 and 12.4 times. Hotspots are enriched for high-impact incidents and align with named blackspots above chance, but cover only 34.6% of high-impact incidents.

Impact-first dispatch was 19.1% worse at realistic fleet load and 23.7–64.7% worse under overload, rejecting H2 and retaining travel distance as the primary dispatch criterion. Forced full diversion increased network total travel time by 18.8 vehicle-hours per scenario, so any future detour policy requires partial compliance and a capacity check.

The ex-DIG's feedback exposed a road-class defect that was fixed. The single structured DMT respondent rated usability at SUS 80.0, while scenario-ranking agreement was inconclusive ($\rho = -0.05$) and therefore only indicative. The official also reported that DMT keeps no timed, location-coded incident records. The defensible present use is an explained triage signal and exploratory placement view, not a calibrated traffic predictor or queue-ordering policy.

9.2 Achievement of Objectives

The deployed score, dashboard, integration and planned experiments were completed. SO5 and the experimental work of SO6 were achieved; SO1 missed end-to-end latency despite fast in-process scoring; SO2 missed H1; SO3 missed

coverage; SO4 produced ranking but not calibrated magnitude; and SO7 obtained only one of the planned five to eight structured responses. Appendix D provides the detailed traceability matrix.

9.3 Individual Research and Product Contributions

The research contributions are an interpretable triage model evaluated under data scarcity, a characterised negative result for impact-ordered dispatch, and a transparent multi-source validation protocol. The product contribution is a deployed, authenticated service with GPS road resolution, dispatch integration and operations views.

9.4 Future Directions

1. Collect timed, geocoded incidents with Police and DMT partners, then recalibrate factors and absolute congestion outputs.
2. Replace kilometre-long slowed lanes with point blockages, multiple seeds, full recovery runs and network-total-delay outcomes.
3. Reweight only from real records or agreement among several experts, including weekday and direction effects.
4. Test partial-compliance rerouting and dispatch policies against independent, demand-sensitive costs.
5. Enable role enforcement, persist score provenance, evaluate road matching, load-test and deploy closer to users.

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GLOSSARY

- **Ablation** — Re-evaluating a model with one factor removed, to measure how much that factor contributes.
- **Blackspot** — A road location with a known history of frequent accidents.
- **Clearance time** — The time from the start of an incident until the road is fully reopened.
- **Hotspot** — A spatial cluster of incidents identified by density-based clustering. In this thesis it is distinct from a police *blackspot*, which is defined by accident frequency.
- **Impact score** — The component's 1–10 measure of an incident's expected traffic disruption, mapped to LOW, MEDIUM, HIGH and CRITICAL bands.
- **Nearest-first dispatch** — Sending the closest available provider to each incident, regardless of its impact.
- **Oracle ordering** — An ordering of the incident queue by the cost model's own predicted delay, that is, an oracle with respect to the evaluation cost. It bounds only the pure queue orderings tested, not mixed policies that combine distance and impact, and it is not an optimal policy.
- **Pre-registration** — Fixing an instrument, analysis and acceptance criterion before any data are collected.
- **Queue ordering** — The order in which a dispatch queue of open incidents is served. Hypothesis H2 tested impact-based queue ordering and rejected it; this is distinct from triage.
- **Rerouting (diversion)** — Sending traffic along a detour that avoids the blocked road segment.
- **Sensitive-location overlay** — A multiplier, capped at +35%, applied after the base score for nearby hospitals, schools, bridges and markets and for long-weekend eves; it is a priority adjustment, not part of the five-factor model.
- **Silhouette coefficient** — A measure, from −1 to 1, of how well each point fits its own cluster compared with the nearest other cluster.

- **Triage** — Classifying an incident's expected traffic impact to inform the type and urgency of response. The thesis rejects impact-based queue ordering of dispatch, not triage.
- **Vehicle-hours lost** — The total extra time spent by all vehicles delayed by an incident.

APPENDICES

Appendix A Research instruments

Two instruments were used with domain experts. The completed and signed forms are held by the author and are available to the examiners on request; they are not reproduced, because the signature and stamp would identify the respondent. The wording of the blank form is given in A.2.

A.1 Interview with the retired DIG (ex-DIG), 27 August 2026. A semi-structured interview with a retired Deputy Inspector General (DIG) of Police (the ex-DIG) covered the existing incident-response pathway, the problem on ordinary roads, the hotspot concept, road classification and temporal patterns of traffic. The findings were recorded in writing on the day. The respondent took part voluntarily and is cited anonymously as personal communication.

A.2 Expert feedback form (5 October 2026). A two-page printed form, completed by a senior Department of Motor Traffic (DMT) official after a live demonstration of the operations dashboard. The session lasted about 40 minutes, of which completing the form took about eight. The consent statement at the top of the form read: "I agree to take part. I understand this form is voluntary and anonymous. It asks only my role and years of experience. Answers will be reported as totals in a SLIIT thesis, and I can stop at any time." The respondent ticked the consent box and signed, dated and stamped the form on 5 October 2026. It had six sections:

1. About you: role and years of experience in traffic or road work (no name).
2. The dashboard: the ten System Usability Scale (SUS) items [45] and two usability items, "The impact score (Low / Medium / High / Critical) is easy to understand at a glance" and "The hotspot map and detour suggestions are easy to read", each on a 1–5 agreement scale. The product has no detour feature, so the second item is read as a rating of the hotspot map (an instrument flaw, Section 7.5). On the returned form, SUS item 10 bore a clear 1 and a crossed-out 5. It was read as 1; had it been read as 5, the SUS score would have been 70.0.
3. Scoring factors: five relevance items, one per factor ("This factor matters for predicting traffic disruption"), and two reliability items ("A score like this would

help police or a control room decide which incident to clear first"; "I would trust the 25 accident hotspots shown, if they were updated with real incident records").

4. Scenario ranking: the ten pre-registered incidents in Table A.1, ranked from 1 (lowest traffic impact) to 10 (highest), without the model's ranking.
5. Clearance time: the model's assumed road-blocking duration for each of seven incident types, with space for the expert's estimate.
6. Open questions: weekday or direction effects, what the system misses for Sri Lankan roads, and which scenarios were hardest to rank.

A.3 Pre-registered scenarios. The ten scenarios were drawn from the scored incidents before contact with any expert, one from each decile of the deployed impact score, with incident type, lane blockage and road class varied. Their identifiers were shuffled so that the order does not reveal the model's ranking. Table A.1 lists them with the model's priority band; Section 6.3.2 reports the ranking result and principal disagreements.

Table A.1: Pre-registered expert ranking scenarios with model priority band

ID	Scenario	Priority
S1	Major accident, 2 of 3 lanes blocked, Prince of Wales Avenue (Sirimavo Bandaranaike Mawatha) (primary), Saturday 08:15	HIGH
S2	Major accident, 2 of 2 lanes blocked, Godagama–Padukka Road (secondary), Tuesday 23:05	HIGH
S3	Engine failure, 1 of 4 lanes blocked, Avissawella Road (primary), Tuesday 06:08	MEDIUM
S4	Overheating, 1 of 1 lane blocked, High Level Road (trunk), Tuesday 15:28	HIGH
S5	Flat tyre, 1 of 3 lanes blocked, Baseline Road (Dr. Danister de Silva Mawatha) (trunk), Tuesday 07:50	HIGH
S6	Overheating, 1 of 2 lanes blocked, Outer Circular Highway (motorway), Wednesday 15:55	HIGH
S7	Minor accident, 1 of 2 lanes blocked, Outer Circular Highway (motorway), Monday 18:13	HIGH
S8	Dead battery, 1 of 2 lanes blocked, Outer Circular Highway (motorway), Wednesday 11:57	HIGH
S9	Out of fuel, 1 of 2 lanes blocked, Malagala Road (tertiary), Sunday 08:07	MEDIUM
S10	Minor accident, 1 of 3 lanes blocked, Galle Road (trunk), Sunday 16:18	MEDIUM

The acceptance criteria, fixed in the protocol before data collection, were a pooled Spearman ρ of at least 0.60 between expert and model ranks across the respondents (a planned five to eight), or a median relevance rating of at least 4 out of 5 for lanes

blocked (CLF) and incident type (ISF), with a written critique of the other factors recorded. The results are reported in Section 6.3.

Appendix B Academic integrity and AI use disclosure

Use of generative artificial intelligence tools

Generative AI tools were used during this research: Anthropic Claude, through the Claude Code assistant, and the Cursor code editor with its built-in AI assistance. They assisted with:

- **Code:** suggesting, writing and refactoring code, and helping to diagnose defects.
- **Analysis scripting:** drafting the analysis, validation and reporting scripts.
- **Figures:** generating the scripts that produce the charts and diagrams.
- **Text:** drafting and editing working documents, instrument materials and the chapters of this thesis.

The affected artefacts, by category, are: the research pipeline (data generation, simulation, validation and analysis scripts); the geo-intelligence service and its automated tests; the geo-intelligence views of the operations dashboard; the figures; the expert briefing and feedback materials; and the text of this thesis.

The author designed the research, made the design decisions, conducted the expert interview and feedback session, and ran and verified every analysis. Every number reported in this thesis was checked against the analysis outputs, and every reference was checked against its source. The author takes full responsibility for the content of this thesis.

Appendix C Research artefacts

This appendix lists the artefacts behind Chapters 3 to 6. The deployed scoring service, its OpenAPI contract and its tests are in the public Kaduna.lk platform repository [4]. The research pipeline and the stakeholder feedback analysis are in the author's private research repository, available to examiners on request; most scripts live under its scripts/ and src/ folders, and the feedback analysis script is kept with the stakeholder kit. Table C.1 lists them in run order. Each stage reads the files

written before it, the analysis scripts do not change the deployed model, and randomness is fixed with seed 42 (the dispatch experiment uses 30 numbered seeds). The pipeline ran with Python 3.14 and the Simulation of Urban MObility (SUMO) 1.26, with its Python bindings (sumolib and TraCI) at 1.27; these and the other package versions are pinned in the research repository's requirements file.

Table C.1: Research artefacts and their outputs

Artefact group	Principal artefacts	Output
Network and incidents	download_colombo_network.py, generate_ incidents.py	Colombo OSM graph; 500 synthetic incidents
Scoring and hotspots	impact_scoring.py, score_all_incidents.py, hotspot_analysis.py	Scores, congestion indices and 25 clusters
Corridor validation	sumo_simulation.py, run_sumo_ validation.py, run_sumo_validation_ coupled.py	Original and demand-coupled SUMO results
Sensitivity and reporting	refine_model.py, validate_weights_cv.py, capacity_sensitivity.py, report_metrics.py	Fitted-weight checks and canonical metrics
Spatial analysis	hotspot_spatial_stats.py, real_data_ comparison.py	Moran's I, Gi*, blackspot and official-statistics comparisons
Dispatch and rerouting	dispatch_baseline.py, dispatch_load_ sweep.py, diversion_experiment.py, sumo_ real_subnet.py	Paired dispatch, load-sweep and diversion results
Transfer and stakeholders	tomtom_anchor.py, generate_expert_ scenarios.py, analyze_feedback.py	Live-speed anchor, pre-registered scenarios and feedback summary
Production component	Geo service, data builders, OpenAPI contracts and tests	Live version 0.2.0, built-in datasets and 146 passing tests

Appendix D Research traceability matrix

Table D.1 links each research question (RQ) and objective (SO) to its method, metric, finding and contribution.

Table D.1: Research traceability matrix

RQ / SO	Method and function	Metric	Result and finding	Contribution / section
RQ1 / SO1	Five-factor service, road resolution, benchmark and tests	Response < 500 ms	p95 9.7 ms in process; one end-to-end call 1.45 s; 146 tests	Deployed artefact / 4.4–6.2
RQ1 (H1) / SO2	120-scenario SUMO grid, ablation, transfer and sensitivity	Pearson $r \geq 0.75$	$r = 0.60$; lanes blocked 0.88; sub-network 0.20: H1 not supported	Interpretable score and validation protocol / 6.1.1–6.1.3
RQ1 / SO4	Point-queue estimator against SUMO	Rank and magnitude error	$\rho = 0.71$ VHL, 0.61 queue; 20× and 12.4× high: relative index only	Model boundary / 6.1.4
RQ2 / SO3	KDE, DBSCAN, spatial statistics and blackspot comparison	Silhouette > 0.3; coverage > 80%	0.64 silhouette; 34.6% coverage; 10 centres near blackspots vs 5 expected	Exploratory hotspot product / 6.1.5
RQ3 (H2) / SO6	30-seed dispatch simulation with independent cost	≥ 20% VHL reduction	19.1% worse at realistic load; 23.7–64.7% worse under overload: H2 rejected	Negative dispatch result / 6.1.7
RQ3 / SO6	96 paired real-subnetwork rerouting runs	Network total travel time	+18.8 vehicle-hours; full diversion worse in all pairs	Rerouting boundary / 6.1.6
RQ4 / SO7	Ex-DIG interview and pre-registered DMT form	SUS, relevance and ranking	SUS 80.0; ranking $\rho = -0.05$; one respondent	Indicative stakeholder evidence / 6.3
RQ1–RQ3 / SO5	Production dispatch and dashboard integration	Geo-sourced production decision	Score 9.2 after version 0.2.0; contract and fallback work	Product contribution / 5.3, 6.4